



08: Detailed Design

IN3065/INM355 | 2025–2026

User-Centred System Design

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Centre for HCI Design | Computer Science | SST

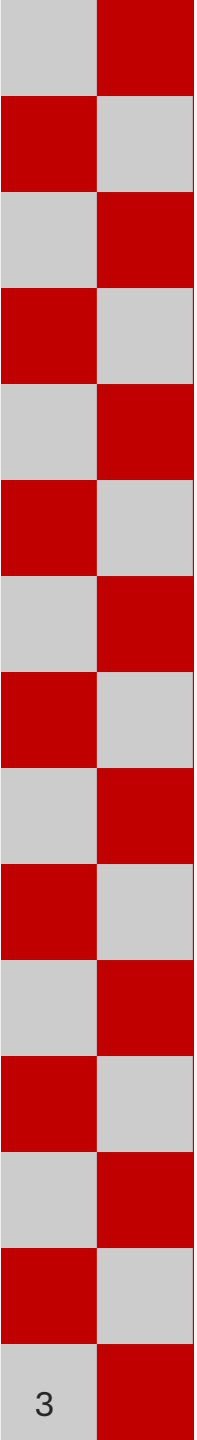
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Session 8: **Learning objectives**



By completing this session, including any practical activity and assigned readings, you should be able to:

- Explain the importance of effective detailed design, and how to achieve it.
- Design usable interactive experiences in detail.
- Document your designs with wireframes, and annotate them, to aid understanding.
- Interpret and apply interaction design principles.
- Explain the use and benefit of design guidelines and design patterns



Detailed Design

(Recap) Conceptual **vs** detailed design

Conceptual Design—*Getting the right design*

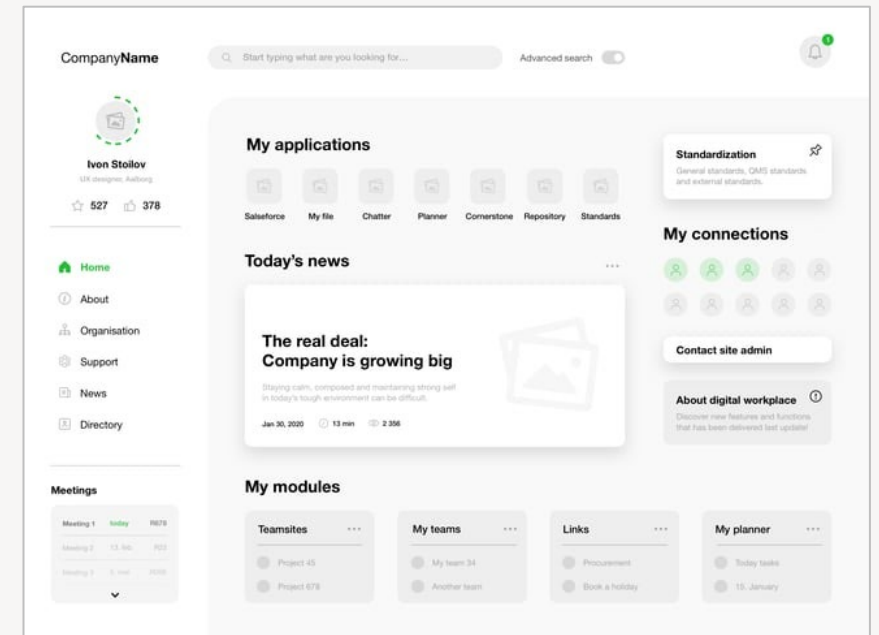
- High-level; focuses on what the system is and does.
- Concerned with users, goals, and context.
- *Defines: what the system will do, interaction types, interface types.*
- Not about screen layout or labels...

Detailed Design—*Getting the design right*

- Screen-level; focuses on how the system will look and behave.
- Defines interface layout, navigation, and visual details.
- Specifies functional logic and interaction flow.

What is detailed design?

- Focuses on specifying *layout, structure, interaction behaviour, navigation, states, content, and feedback* of an interactive system.
- Translates concepts / conceptual models, flows, and requirements into concrete interaction design specifications.
- Gradual progressing and refining of ideas.
- Moving from ‘what’ to ‘**how**’.
- Lots of decisions to be made!



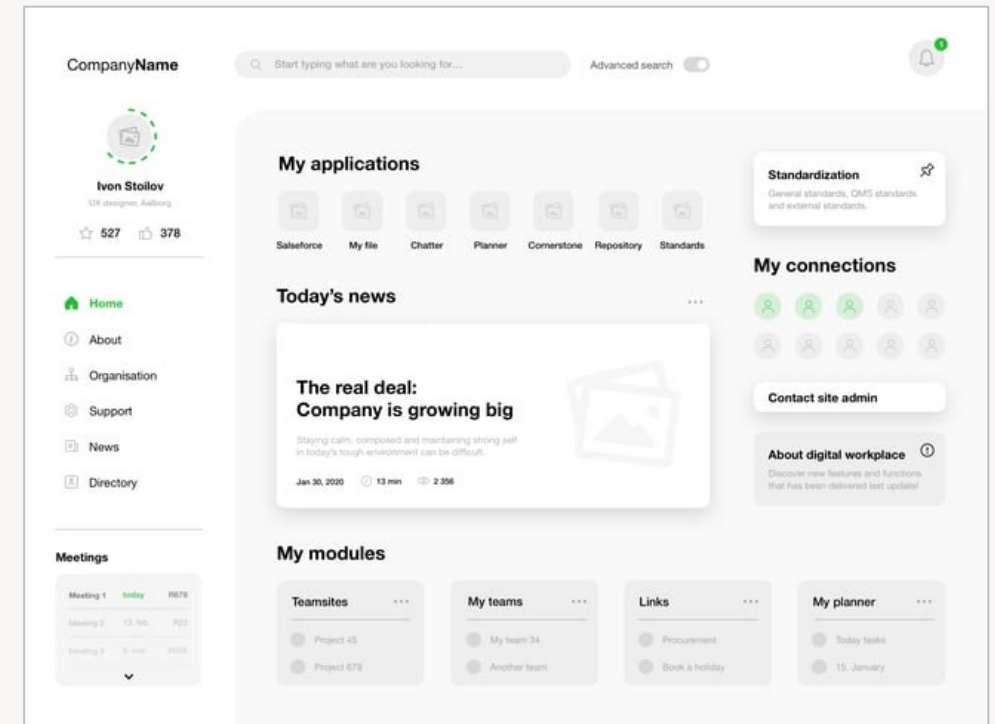
Why detailed design matters

- Ensures that designs are testable and buildable
- Aligns design, product and engineering/development
- Surfaces edge cases and inconsistencies
- Supports usability testing
- Prevents premature focus on aesthetics

From concept **to detail:** decisions, decisions...

Things to address / figure out:

- Core concepts
- Tasks (& mental models)
- Flows
- Information hierarchy
- Wireframes
- Behaviour & states
- Interaction rules
- Visual design (later)





Detailed design / Flow diagrams

Wireflows

(Recap) Flow diagrams

- Visual representation of **flow** through a system when performing a particular task—the sequence of **steps** or **actions** a user goes through to achieve a goal.
- Help work out the **logic of interaction and behaviour**, including the **paths** through the system, before design—make sure goals and interaction are properly **supported**.
- **Important:** not a map of all functions or routes in a system—concerned with a particular task/goal

Example: Simple user flow

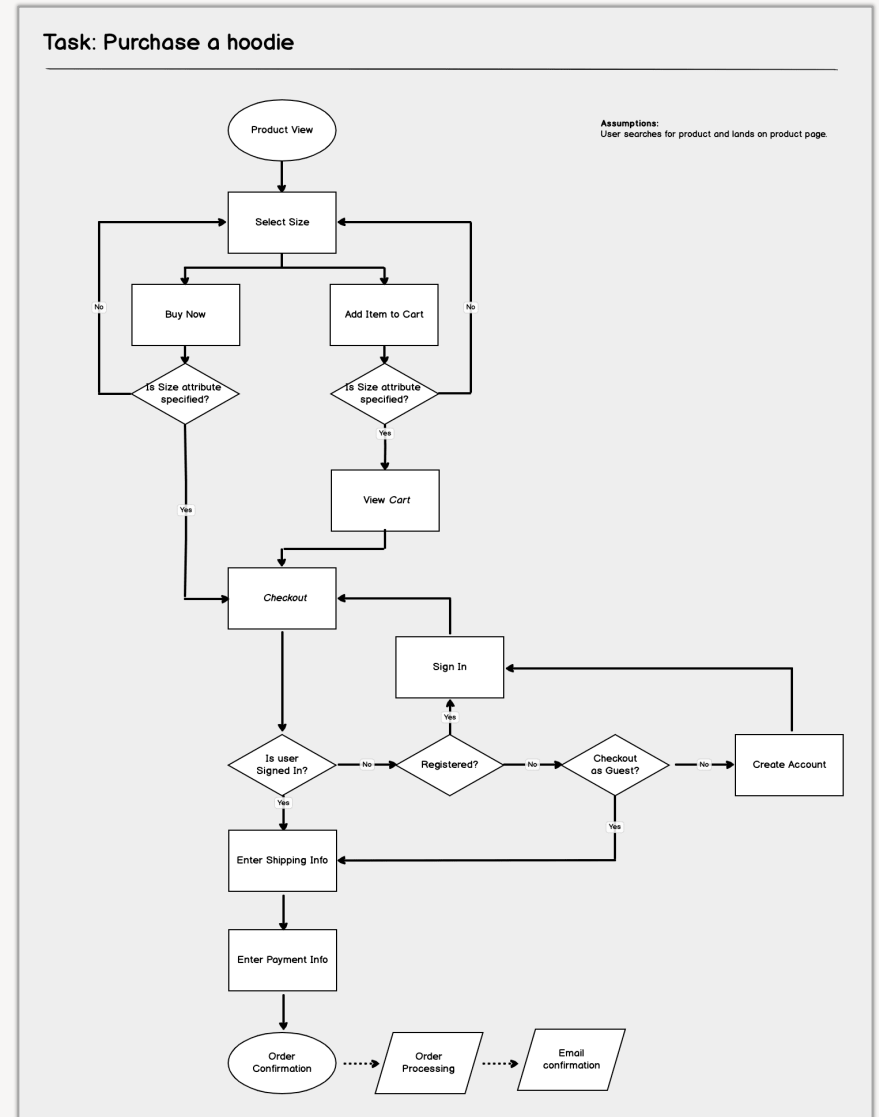
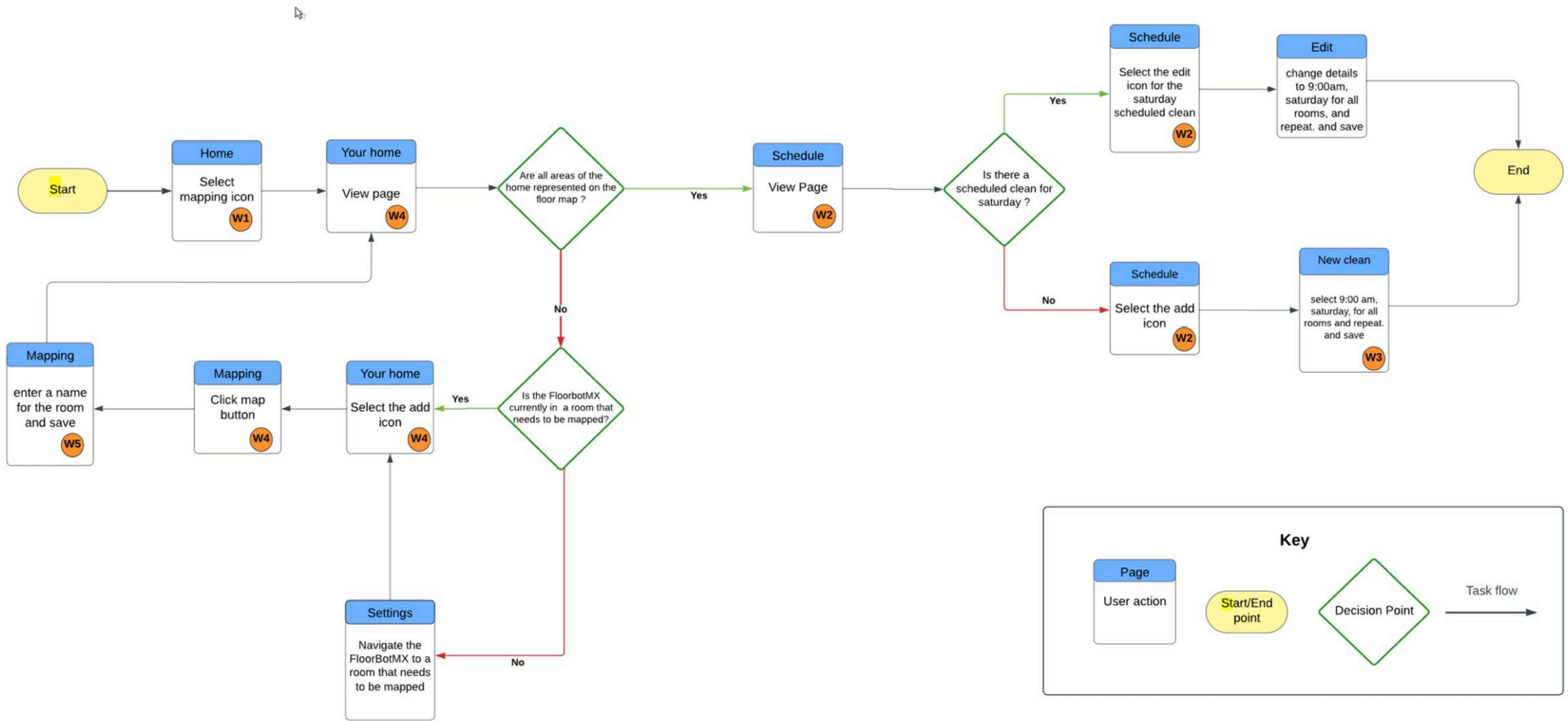


Image from <https://balsamiq.com/learn/articles/wireflows>

(Recap) Example: User flow

Note: Can see which page/screen an action is happening on; useful cross-referencing to relevant wireframes, by using their IDs (W1 etc). The goal/task is missing though

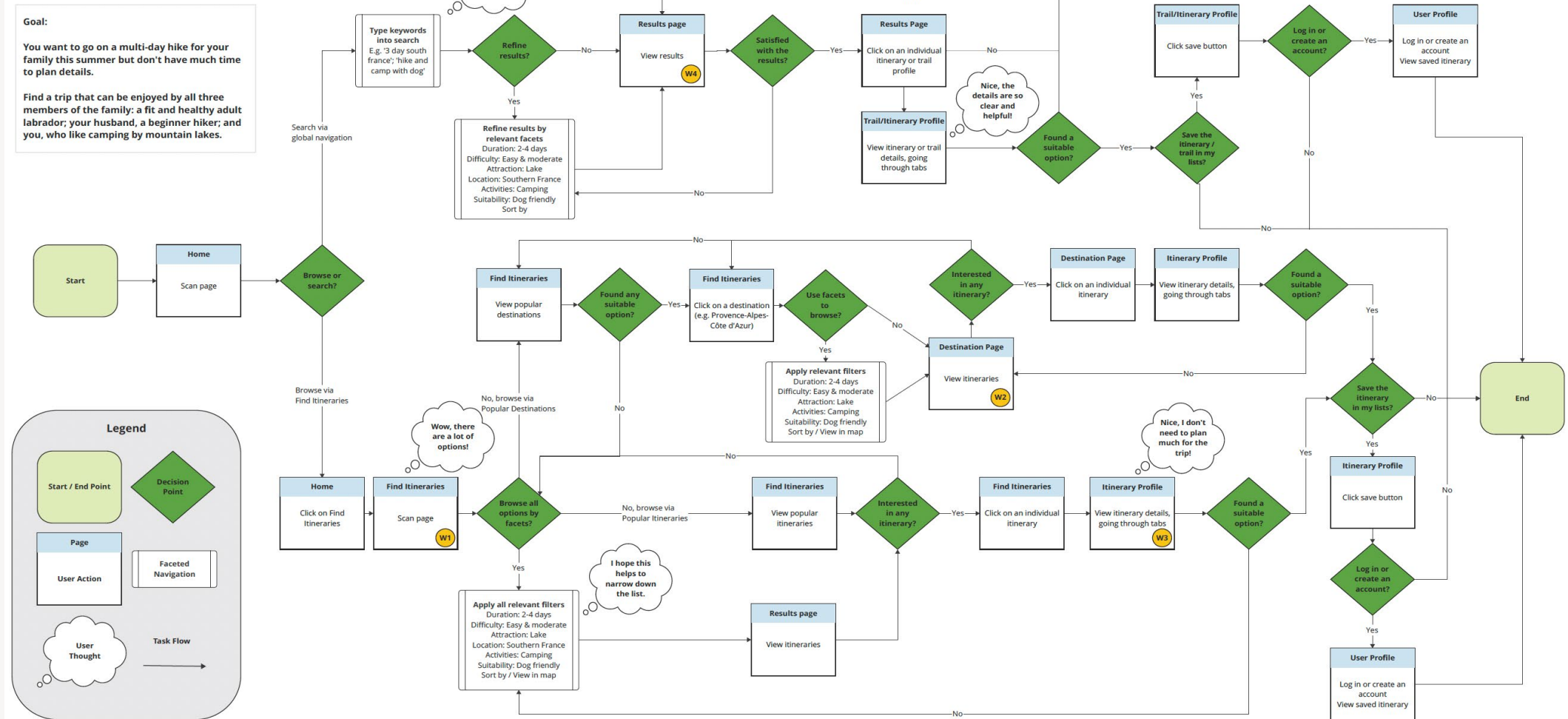


Created by previous student

(Recap) Example: User flow

Note: Notation choices aid the reader's understanding

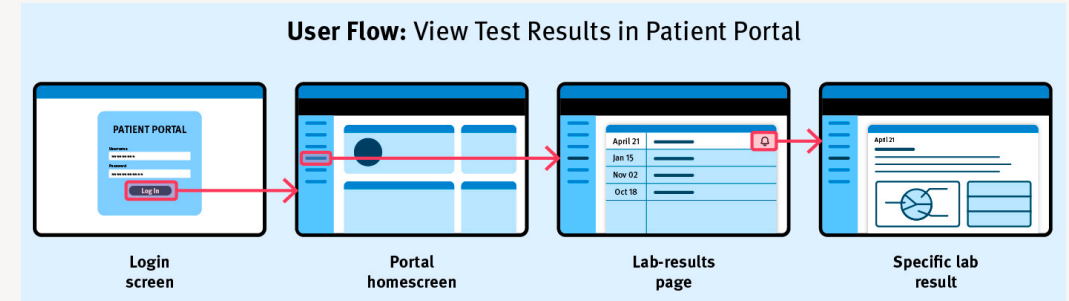
Appendix 3: User journey



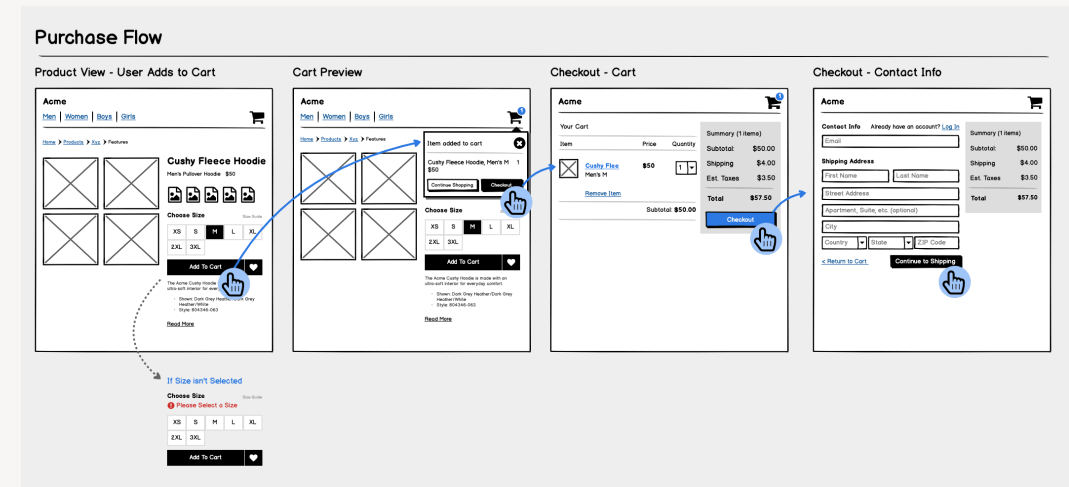
Created by previous student

Wireflows / Screen Flows

- Task / user flows can also be represented as wireflows / screen flows.
- Typically combine task/user flow and wireframes (or higher fidelity designs)
- Visual illustration of pathways through a system, including screen transitions.
- See: <https://www.nngroup.com/articles/wireflows>

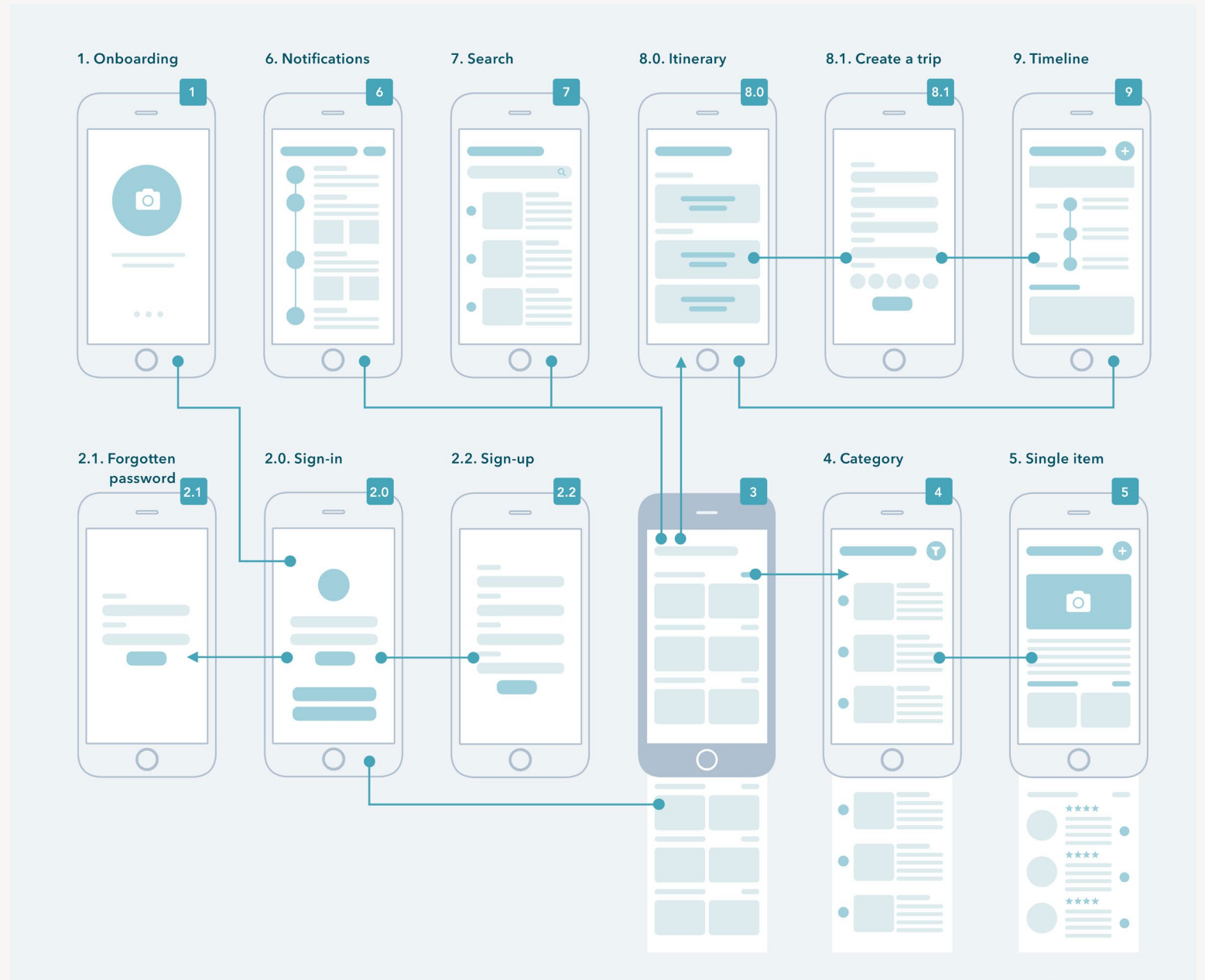


From: <https://www.nngroup.com/articles/user-journeys-vs-user-flows>



From: <https://balsamiq.com/learn/articles/wireflows>

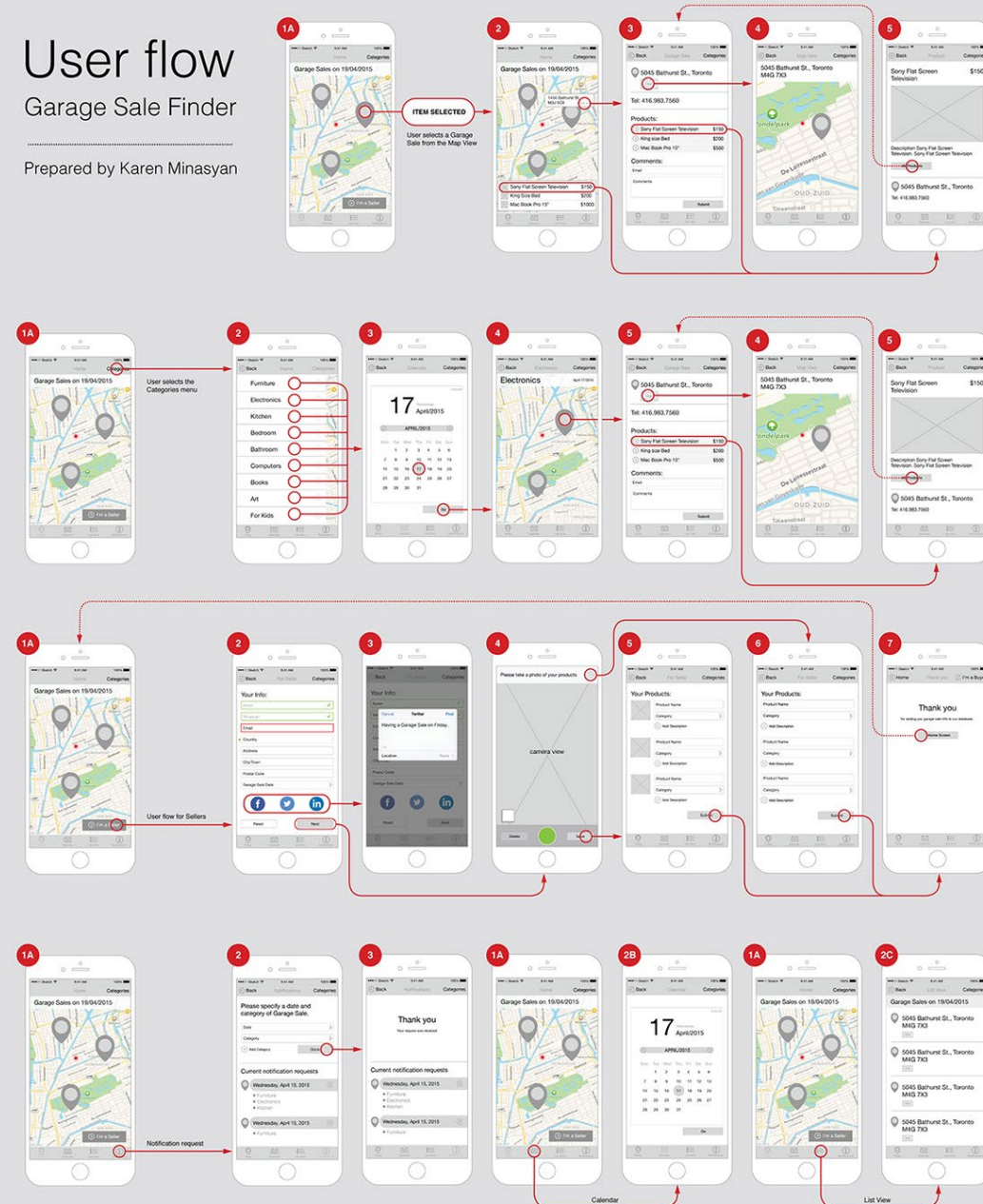
Example: Wireflow



Example: Wireflow

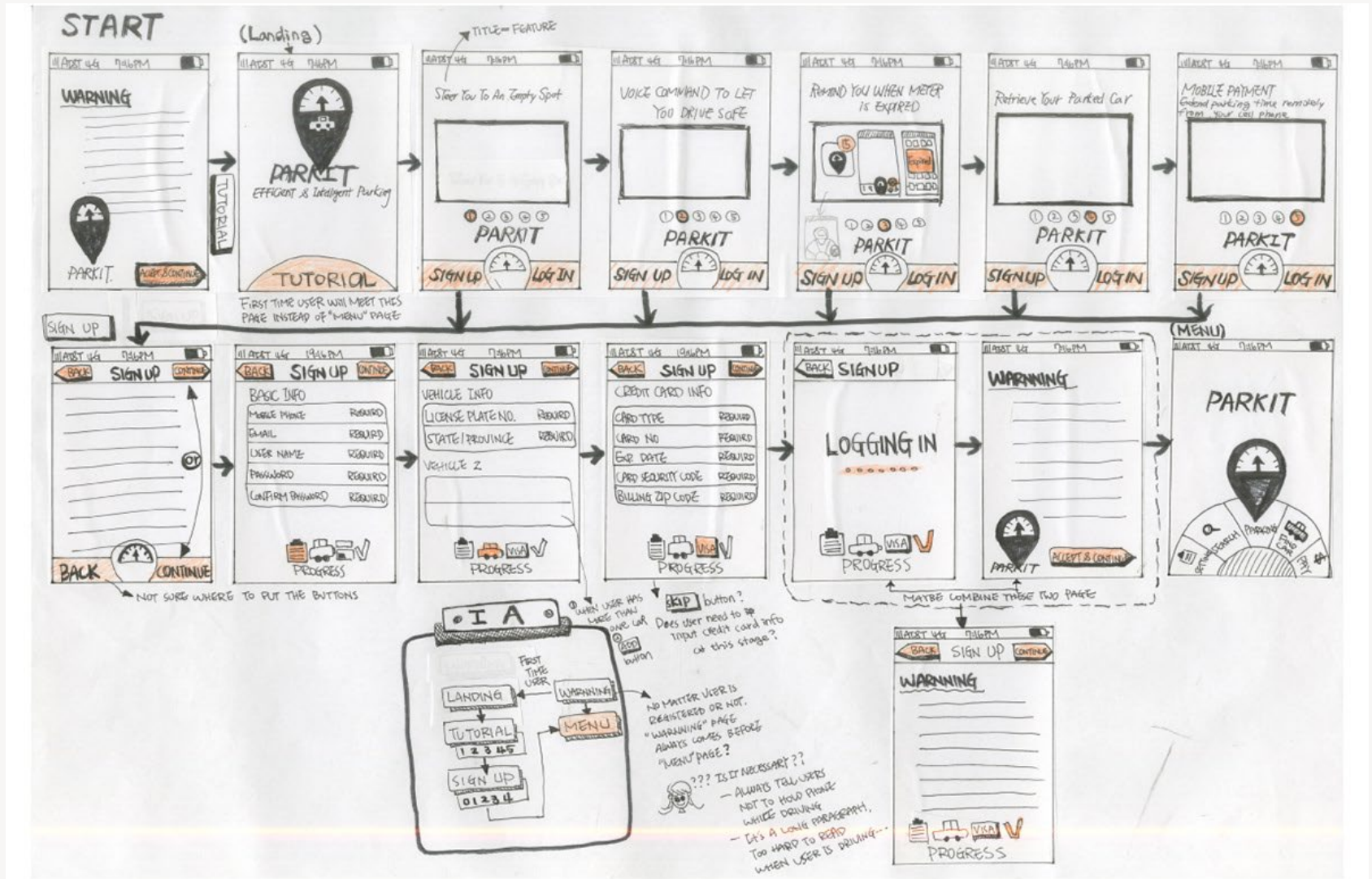
User flow Garage Sale Finder

Prepared by Karen Minasyan



From: <http://www.kmdesignhouse.com/ux.html>

Example: Wireframes / Wireflow



(Recap) Flows within a design process

- Useful for working out the **flow of interaction**
- Useful for determining/planning what **screens/interactions** need to be designed—e.g. before wireframing.
- Useful for spotting **gaps/problems** in a system—when planning a new system or analysing an existing one.
- **Wireflows** are useful for mapping out the flow of screen/page transitions and changes, triggered by interaction with specific elements—usually when you have some wireframes created.
- Useful collaboration tool—aid understanding

CW2 coursework User Flow deliverable

- Pay close attention to the coursework brief.
- See the lecture 7 slides for more tips / watchpoints.
- If you have questions, ask.



Questions?





Detailed design **Wireframes**

(Recap) Wireframes

- Two-dimensional **representation of an interface**.
- Simplified view of what will appear on a screen/page.
- They operate as **blueprints/schematics**—show elements and their locations.
- Concerned with
 - **Structure** and **layout** of screens/pages,
 - Available **functionality** (including navigation)
 - What a user can **interact** with and what happens when they do.
- Usually don't contain colour/typography/images.
- **Annotations can provide additional detail**

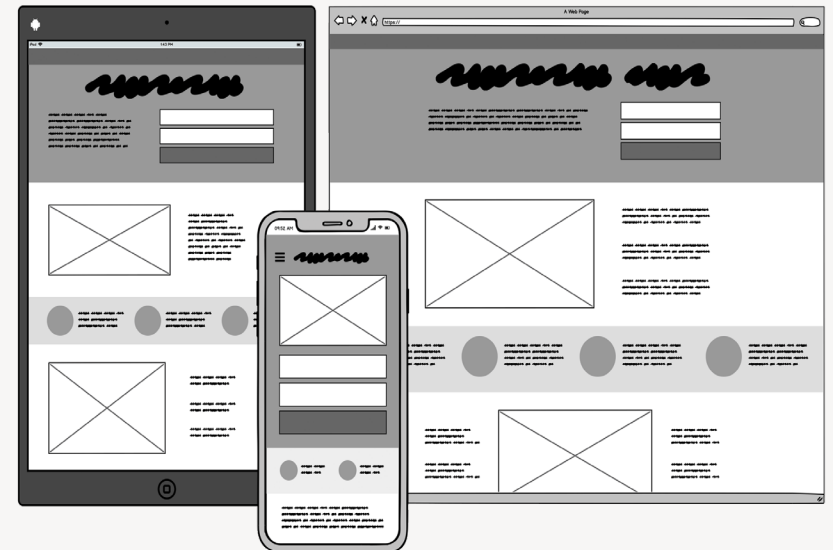
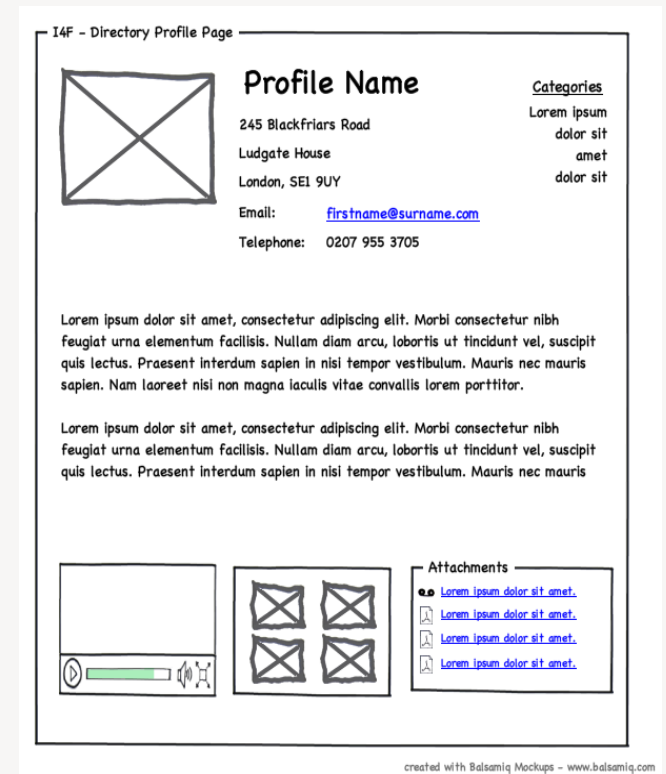


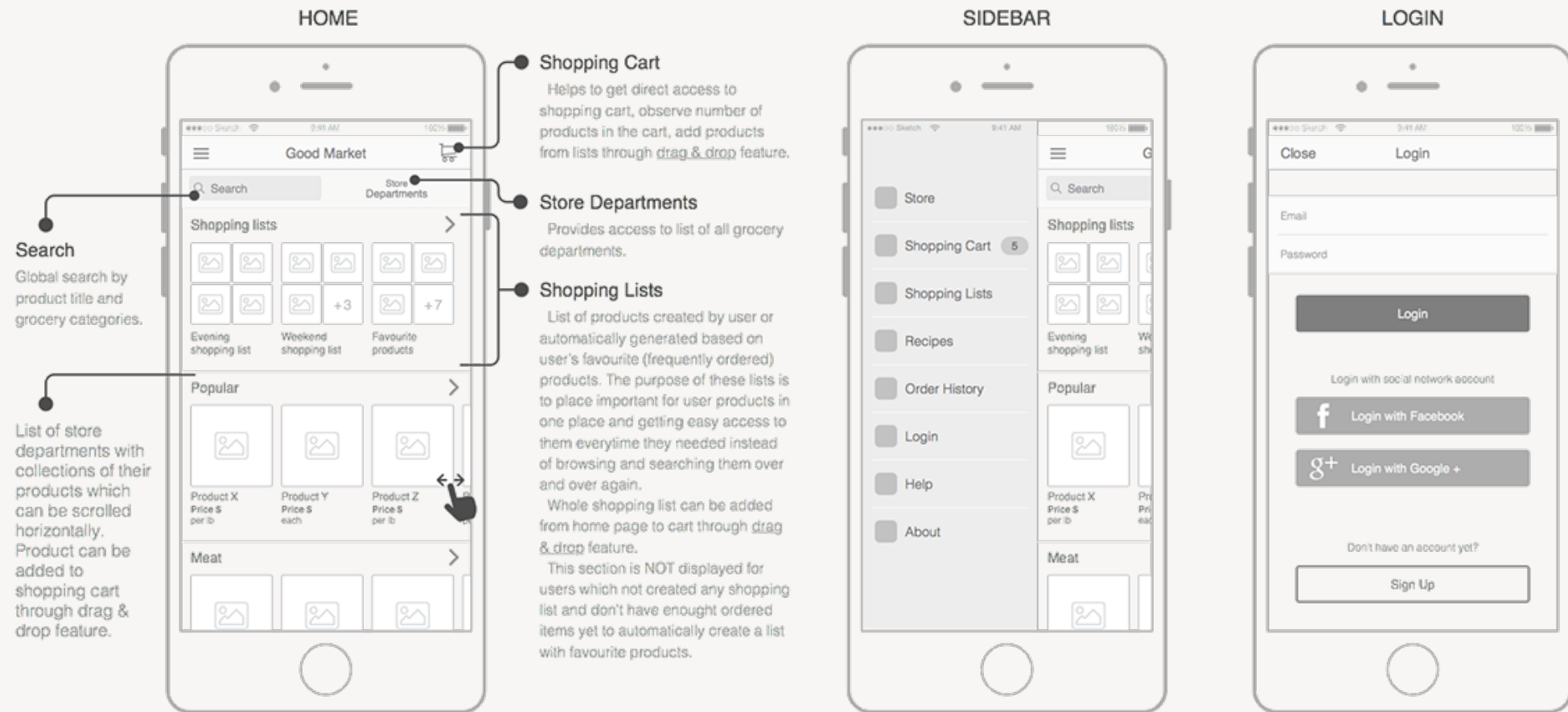
Image from: <https://balsamiq.com/learn/articles/what-are-wireframes>

(Recap) Why do we **need/use** wireframes?

- Wireframes are used to:
 - Think through the structure of screens.
 - Organise interface elements and navigation.
 - Communicate design ideas to stakeholders and team members.
 - Represent screens that support interaction flows, in evaluation activities or interactive prototypes.
- In later stages of design, wireframes may become more detailed design artefacts, working as *specifications*—more about that today



Wireframes example



From: <https://www.behance.net/gallery/32614399/UX-Wireframes-for-Online-Grocery-Shopping-Mobile-App>

Creating a wireframe

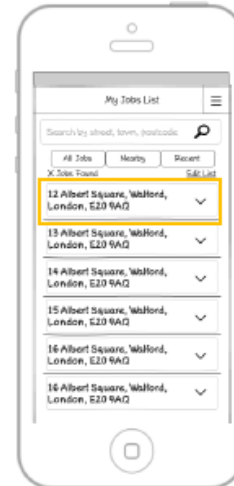
- Use your flows in conjunction with conceptual design thinking / outputs.
 - What are the main features, objects and actions?
 - What needs to happen or be available?
- Think about objects and structure / information architecture.
 - What needs to be on the page and in what format?
 - How do these elements relate to each other?
 - How should they be prioritised on the page?
- Observe good practice.
 - Follow design principles / guidelines / conventions / patterns.
 - Use a grid structure.
 - Clarity is key.

Start by sketching



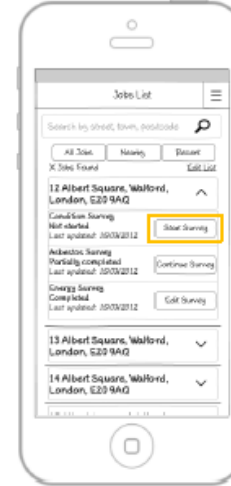
City Interaction Lab

Human Centred Innovation



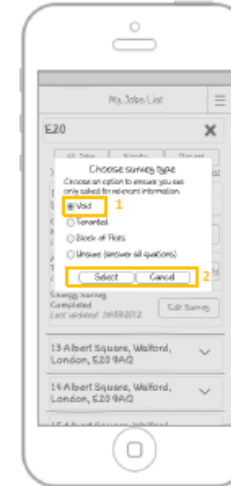
My Jobs List

On launch surveyors are presented with a list of properties assigned to them. This can be filtered in real time using search or sorted using nearby / recent buttons.



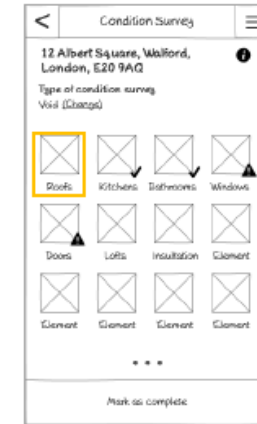
Start condition survey

Clicking on a property expands it to show the applicable surveys with status information and buttons to begin surveys.



Choose survey type

On this overlay the surveyors are presented with labelled radio buttons where they can choose a particular condition survey type to complete. A label explains the benefit of choosing the survey type. If more than 4 options are available this list should be displayed as a dropdown menu. An unsure option is included which then presents surveyors with a survey including all of the questions. This list could have a smart default option selected based on the survey type most frequently used. Cancelling will return to jobs list screen.



Select element

Starting the condition survey the surveyor will be presented with a screen displaying all of the applicable elements. Elements will have an indicator showing whether they are partially or fully completed. More elements can be accessed by swiping the page. Surveyors can access household information via the (i) icon at the top of the page.

Once surveyors feel that they have completed all the necessary elements for a property they can mark this element as complete.

Using the back button returns to jobs list.



Sub element grid

Once surveyors have chosen an element from the first grid view they are presented with a similar grid view for all of the sub elements to survey. Surveyors can complete this in any order they want.

Once surveyors feel that they have completed all the necessary sub elements for a property they can mark this element as complete.

Using the back button shows this element as partially complete on previous grid.

Promaster Mobile – User Journeys – Condition Survey: Navigate to sub element

Approval

This document is a project deliverable. By signing the cover sheet you are confirming receipt of the deliverable and agree to proceed with the project based on the contents.

Version 1.0

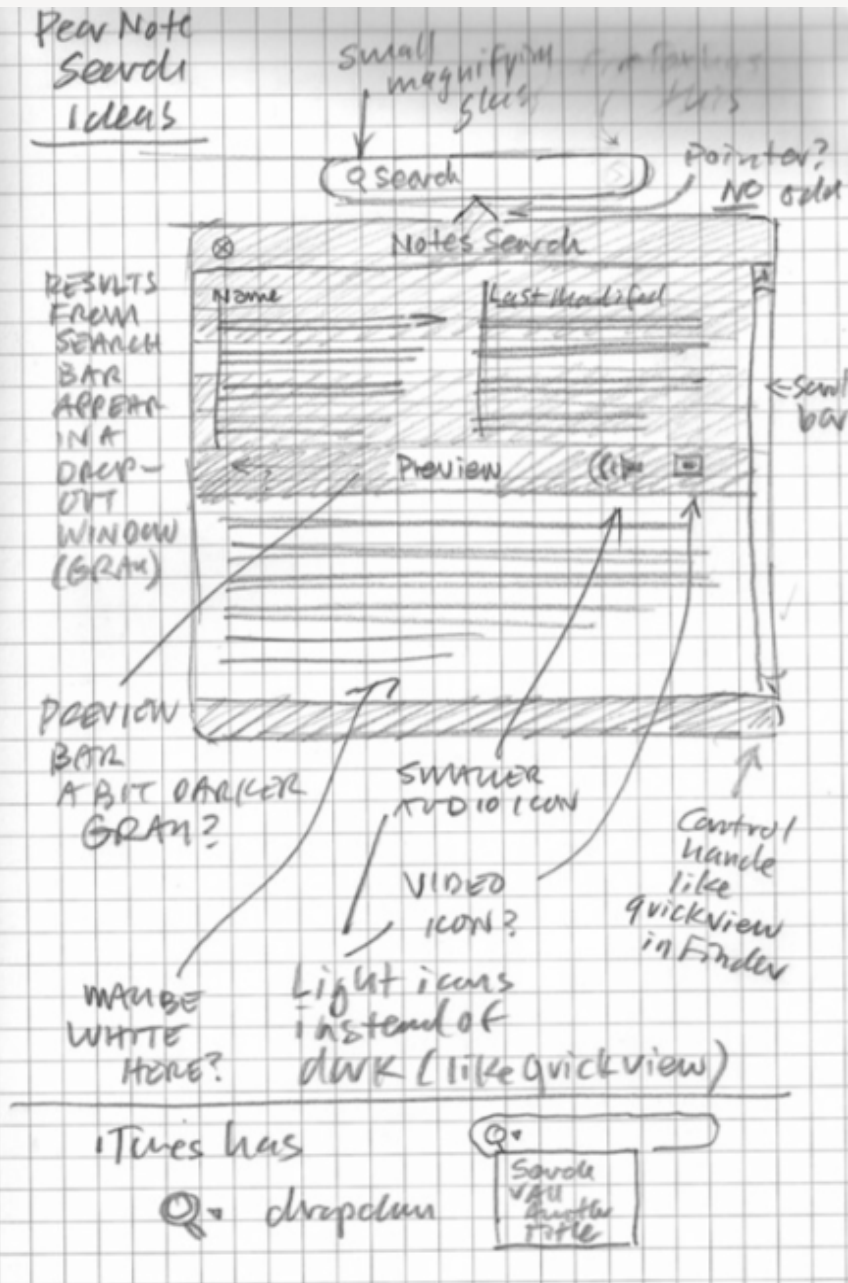
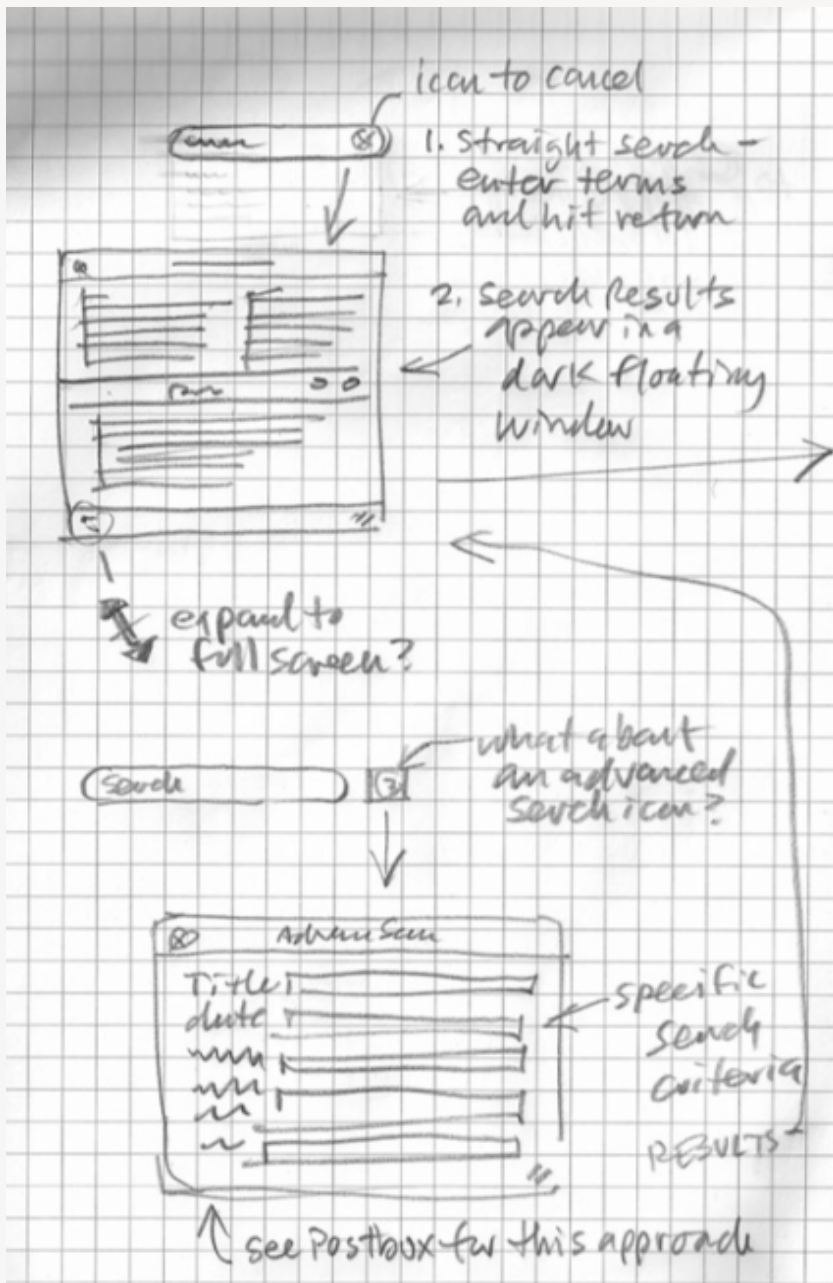
05/08/2014

By: Stuart Scott

For: Andrew Squires

Page 3 of 9

From: City Interaction Lab



Wireframes: Basic elements

— Fr

Placeholder...

Filled

Disabled

Search...

Select...

Selected

Select...

Placeholder...

Placeholder...

BUTTON 1

BUTTON 1

BUTTON 2

BUTTON 2

- ☐ Radio
- ☒ Selected
- ☐ Disabled
- ☐ Disabled
- ☐ Checkbox
- ☒ Checked
- ☒ Disabled
- ☐ Disabled



Label

[Links](#)



Modal

×

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pretium pretium tempor. Ut eget imperdiet neque. In volutpat ante semper diam molestie, et aliquam erat laoreet. Sed sit amet arcu aliquet, molestie justo at, auctor nunc.

SECONDARY

PRIMARY

Heading 1

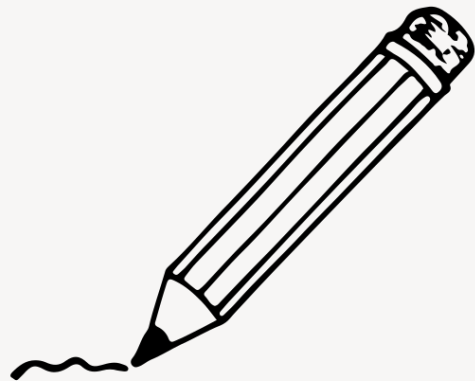
Heading 2

Heading 3



Tools for designing wireframes

- Draw by hand — definitely to start with at least!
- Lots of software tools available, some specialist and others multipurpose. Many include wireframe templates, component libraries, etc.
 - e.g. Figma, Axure RP, Balsamiq, Penpot, Miro, InVision, Sketch, Lucidchart...



axure

balsamiq®



Figma



Sketch



penpot



Lucidchart



miro

Good wireframes include

- Clear screen title
- Content hierarchy
- Interaction elements
- Navigation
- Components & annotations
- States: empty, error, loading, success
- Behaviour notes
- Constraints (disabled, validation)

Detailing interaction **behaviour**

- Interaction behaviour defines how the interface behaves, and responds to user actions.
 - **Action-response rules**: What happens when
 - **State transitions**: idle → loading → success/error
 - **Feedback** (e.g. visual, auditory, haptic)
 - **Constraints** (what's allowed / not allowed)
 - **Signifiers & affordances**: make actions discoverable
 - **Recovery actions** (undo, cancel, retry)

System states

- We should design for all key system states
 - **Idle:** default/starting point
 - **Loading:** progress indicators
 - **Empty:** no data yet, first-time use
 - **Partial:** limited/incomplete data
 - **Error:** failures, validation issues
 - **Success:** confirmations, transitions
 - **Unavailable:** permissions, offline
 - **Transient:** saving, processing

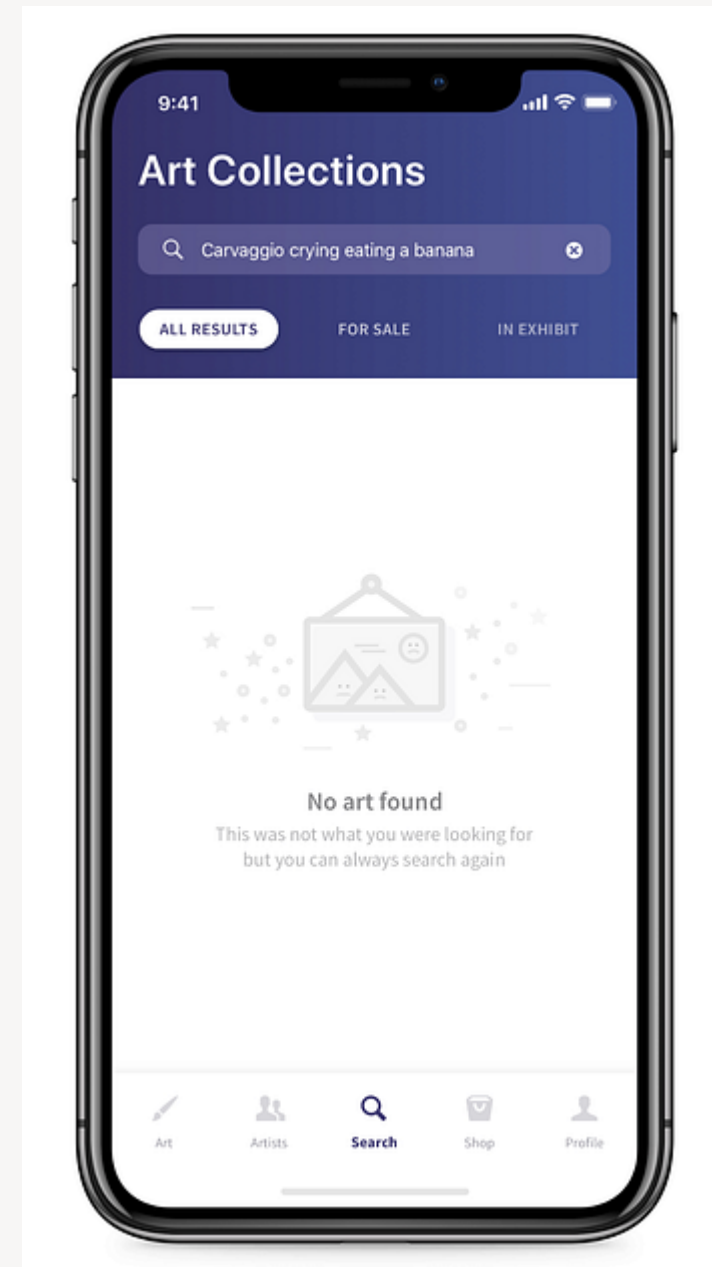


Image source: <https://uxdesign.cc/designing-for-different-ui-states-87d60130f85f>

Error handling and constraints

- **Inline validation:** Non-modal; contextual
- **Prevention:** Constraints, e.g. disabled states, input masks, defaults...
- **Actionable messages:** What the user should do
- **Clear recovery paths:** e.g. undo, cancel, retry

Image source: <https://www.nngroup.com/articles/error-message-guidelines>

The screenshot shows a web form titled "CLEAR" with the subtitle "Change Your Password". The form is enclosed in a light gray border. At the top, it says "Enter a new password for [redacted email address]". Below this, it says "Make sure to include at least:". There are four criteria listed with checkboxes: "8 characters" (checked), "1 uppercase letter" (unchecked), "1 lowercase letter" (checked), and "1 number" (unchecked). There is also a "1 special character" (checked) with an information icon. Below the criteria, there are two password input fields. The first is labeled "* New Password" and has a red border with the text "Too weak" on the right. The second is labeled "* Confirm New Password". Below the input fields is a "Change Password" button. At the bottom, it says "Password was last changed on 2/9/2023 6:29 PM." and a copyright notice "© 2023 clearme.com. All rights reserved."

Information Architecture and navigation

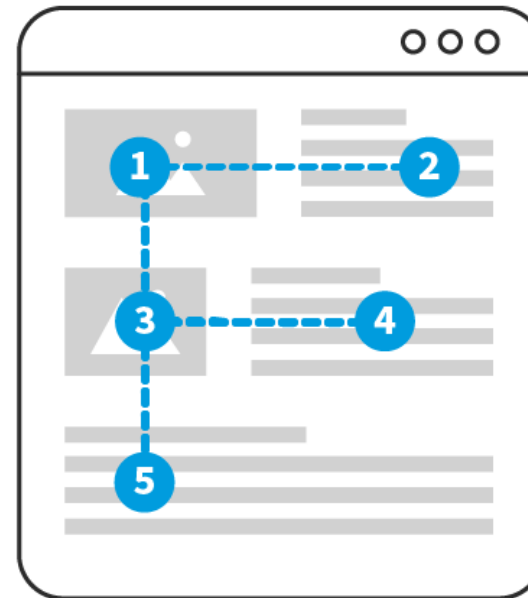
- Choose a **navigation model**:
 - E.g. top navigation / sidebar / tabs / back stack
- Navigation should **orient**
 - Where am I? Where can I go from here?
How can I get back?
- Keep it **simple and consistent**:
 - Avoid deep nesting unless there's a clear reason.
 - Make “home” and “back” predictable.
 - Put things where users will expect to find them.



Information **hierarchy**

- Primary vs secondary content/information
- Chunking & grouping
- Visual weight: size, contrast, spacing
- Scannability / Scanning patterns: F-pattern, Z-pattern
- Progressive disclosure: reveal detail when needed

F-Pattern



Z-Pattern

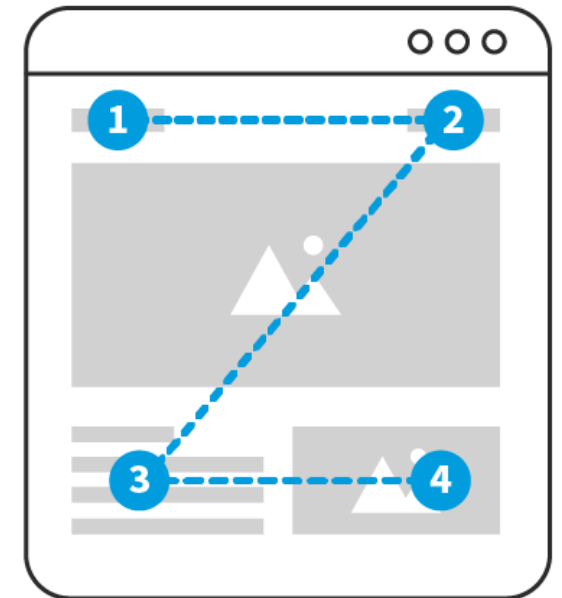


Image source: <https://www.interaction-design.org/literature/topics/visual-hierarchy>

Layout decisions

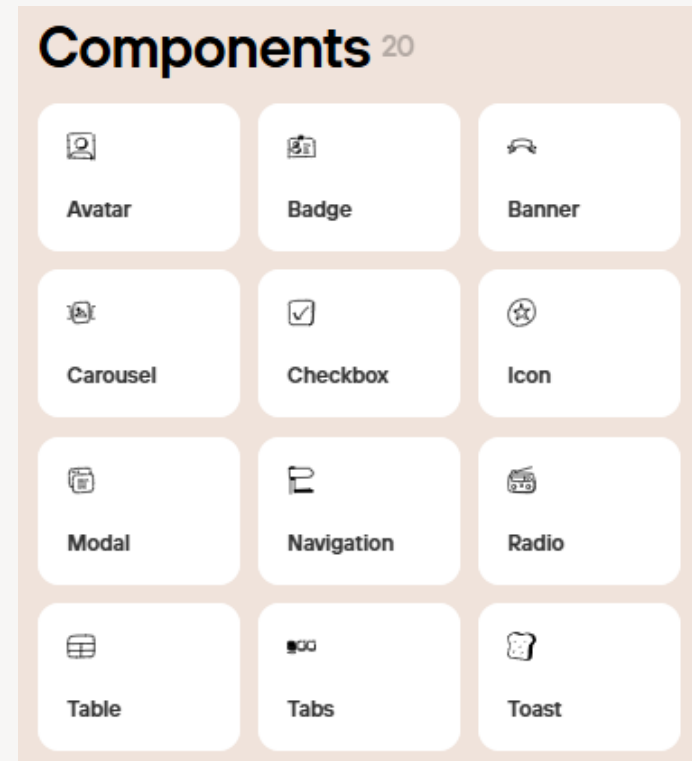
- **Grouping & alignment**: create structure and clarify relationships.
- **Spacing & whitespace**: reduce cognitive load, increase legibility.
- **Density vs clarity**: balance information richness with readability.
- **Flow direction**: guide attention via vertical/horizontal sequences.
- **Responsiveness**: adapt layout across screen sizes and contexts.



Image source <https://www.interaction-design.org/literature/topics/negative-space>

Design Patterns

- Solutions to recurring problems, validated in practice.
- Common examples:
 - Date pickers
 - Card layouts
 - Accordion (progressive disclosure)
 - Infinite scroll vs pagination
 - Faceted filtering (checkboxes, sliders, dropdown filters)
 - Pull to refresh
 - Navigation bars
 - ...



Questions?





Detailed Design

Wireframe annotations

Wireframe **annotations**

- Provide additional information about a design that can/will aid understanding of it, typically in the form of **callouts**.
- Useful for clients, developers, visual designers, team members, other stakeholders, copywriters, future self, etc...
- **Pre-empt (answer) questions** such as
 - What is the purpose of this section?
 - What will happen if a user clicks on this button?
 - What will happen if a user hovers their cursor over this?
 - What will the user see if they are/aren't logged in? If they enter invalid data?
 - If a user clicks this link, where will they be taken?
 - etc...
- Think in terms of content, functionality & behaviour—**what do you want or need to explain?**

Wireframe annotations: **what & when?**

- Explain interactive behaviours
- Describe what dynamic elements do.
- Show call to actions (CTAs).
- Clarify elements and sections.
- Explain links and buttons.
- Show active and inactive states for page elements
- Explain content that is hidden, conditional or modal.
- Provide supportive messaging and prompts.
- Provide error messaging and error states.
- Indicate page and element transitions.
- Provide notes for the project team
- Explain design decisions
- ...

See this week's readings
on annotations

Wireframe annotations: **how?**

- Typically, adjacent to the design.
 - Don't force the reader to have to flip back and forth between pages!
- Usually in text form.
 - Although may also include additional images.
- Different ways to represent and present them
 - Numbers, lines, graphics, etc.
 - Text boxes, tables, side panels.

Wireframe annotations example

Payment Details Pinned Tray

The wireframe shows a 'Payment Details Pinned Tray' with the following elements and annotations:

- 01** Close Payment Details Button: A green checkmark icon with the text 'Close Payment Details'.
- 02** See prices with sales tax: A checkbox.
- 03** See prices with delivery and destination fees: A checkbox.
- 04** Please select a Province: A dropdown menu showing 'Ontario'.
- 05** Available Rebates: A heading.
- 06** \$500 Hyundai Military Rebate: A checkbox with a green checkmark.
- 07** \$750 Hyundai Loyalty Rebate: A checkbox.
- 08** \$350 Hyundai Graduate Rebate: A checkbox.
- 09** \$14,000 Ontario Electric Vehicle Incentive Program: A checkbox.
- 10** Legal: A link.

The tray contains three cards: Cash, Finance, and Lease. Each card displays a total price, estimated trade-in value, and a list of items with their prices. The Cash card shows a total price of \$30,429. The Finance card shows a total price of \$390/month. The Lease card shows a total price of \$454/month. Each card has a 'SELECT' button.

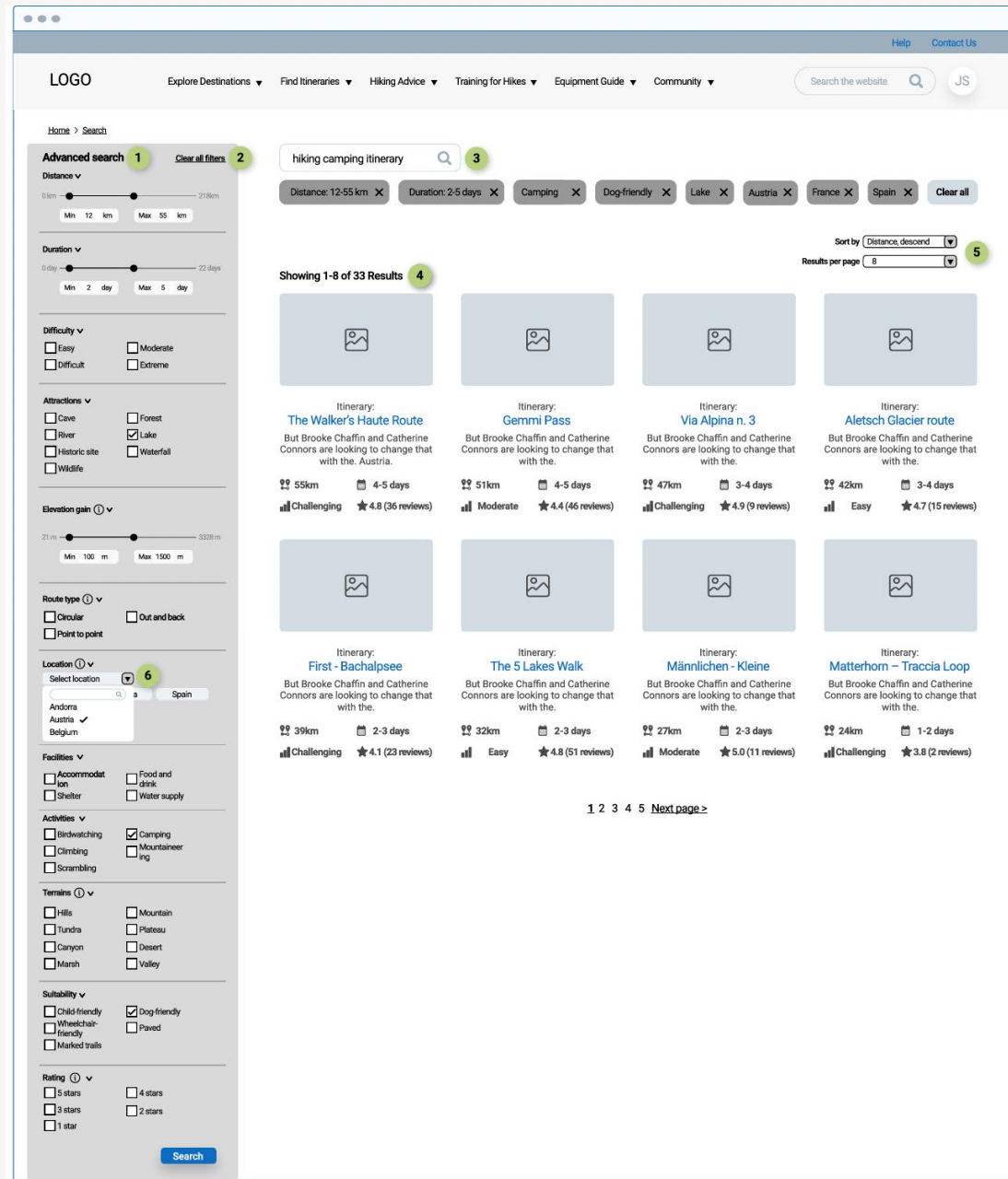
NOTE: (AR) = Accessibility Requirement

- 01 Close Payment Details Button**
 - On click, collapses the tray.
 - (AR) For screen reader users, the screen reader should say something like "[Button] close payment details".
 - (AR) Make the chevron, the circle surrounding the chevron, and the text "close payment details" as one clickable item.
- 02 Sales Tax Optional Checkbox**
 - This is an optional checkbox.
 - On selection, a line item is added to each of the 3 cards (cash, finance, lease) that states the sales tax.
- 03 Delivery and Destination Optional Checkbox**
 - This is an optional checkbox.
 - On selection, a line item is added to each of the 3 cards (cash, finance, lease) that states the fee.
- 04 Dropdown**
 - This dropdown displays the user's selection. The user would have selected the province prior to opening the drawer for the first time.
 - (AR) Please follow accessibility best practices for dropdowns.
- 05 Heading**
 - (AR) This is a heading level 3 (assuming this drawer is made a landmark when in its collapsed state and as such it has a hidden heading level 2 as stated in the annotations for step 1).

From: <https://xd.adobe.com/ideas/process/wireframing/benefits-of-annotating-wireframes>

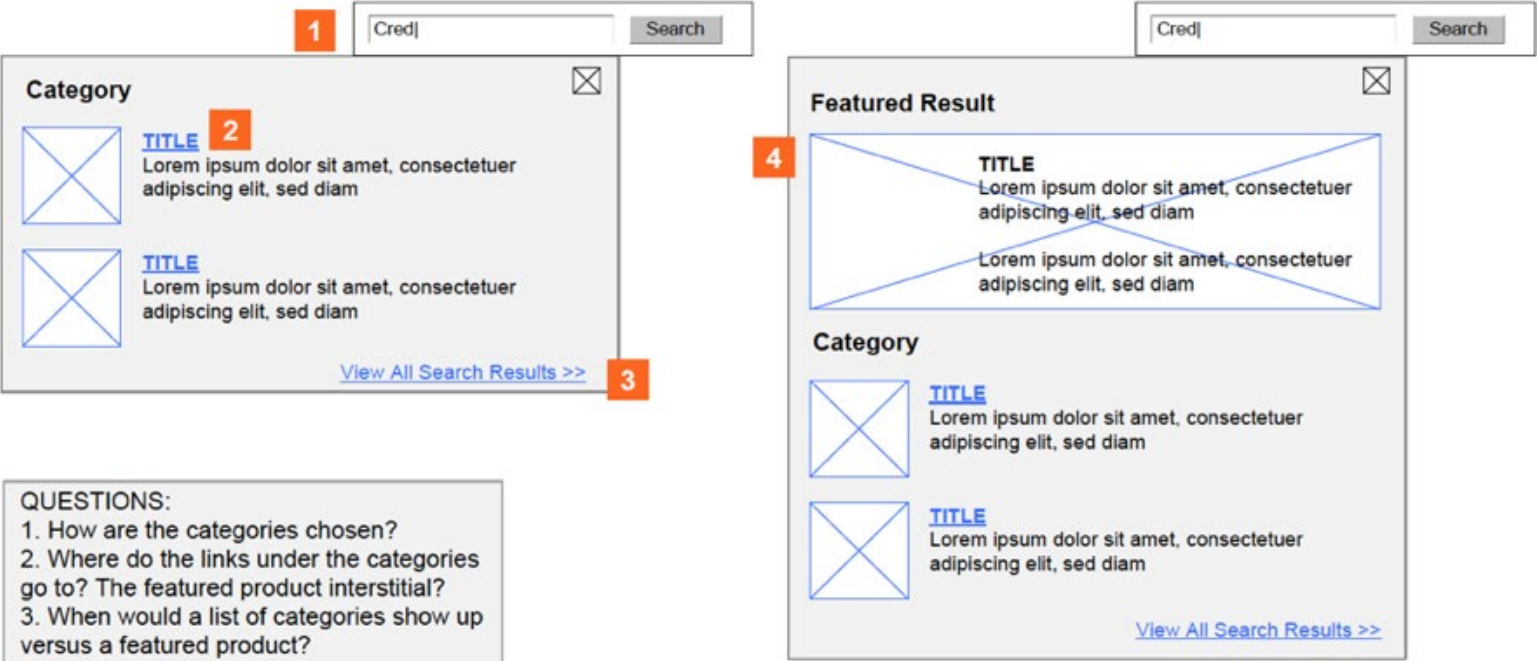
Wireframe annotations example

Note: Annotations are not as comprehensive as we would expect for the CW2 coursework—some interactive items are not annotated, which means important information is missing for anyone wanting to fully understand these designs.



- 1 The user can set filters by using the checkboxes, sliders and list, clicking on 'Search' to see results
- 2 The button serves to clear all filters
- 3 The system supports matching keyword variations in search to maximise results. It also automatically recognises misspellings and informs the user with a hyperlink to the exact term on top of the list of results from close variants.
E.g. The user enters 'tiroll' and the results for Tyrol show with a message on top: 'We searched for Tyrol, France. Did you want to search tiroll?'
- 4 When there are no matching results, the zero-result message will be immediately followed by a row of related links, providing a safety net to avoid dead ends.
E.g. The user enters 'pack 2kg bag 3 day hike Austria winter'. The message suggests alternatives: 'Try winter hike Austria?' with the next row recommending related content with hyperlinked gallery: 'You might be interested in Ultralight backpacking checklist'
- 5 The user can change results per page and sort results by distance (ascending/descending), duration (ascending/descending), rating (high to low/low to high), time published (oldest to newest/newest to oldest) and alphabets (A to Z/Z to A)
- 6 A scrollable multiselect dropdown list (the sidebar showing scroll needs to be added); selected items show as tags under the list

Wireframe annotations example



QUESTIONS:

1. How are the categories chosen?
2. Where do the links under the categories go to? The featured product interstitial?
3. When would a list of categories show up versus a featured product?

Notes			
Label	Name	Annotation	Source
1	Search Response	Display and update search suggestions as the user enters keywords.	Instant feedback will help users refine their search and get great results without loading any new pages.
2	Image/Text Link	Clicking on the link will take you to the search results page.	None.
3	View All Search Results	Clicking on the link will take you to the search results page.	None.
4	Featured Result	Pictures are more memorable than a string of text and can give important cues about the nature of the content. Clicking on the link will take you to the featured result on the search results page.	Business owners often compete for real estate and users' attention on the home page. Give business owners the opportunity to feature their products or services each time a user enters a specific keyword.
5	Category Results	Implement simple type-based categorization. Clicking on a link goes...	Help users instantly weed out content that they're not interested in. "I don't want a new card; I want information about benefits."

Wireframe annotations example





Human Centred Innovation

Text introduction explaining what samples and quotas are and the relationship between complexity and cost.

Users must provide a sample name. This can be used for later reference.

Slider allows users to choose their sample amount with granularity.

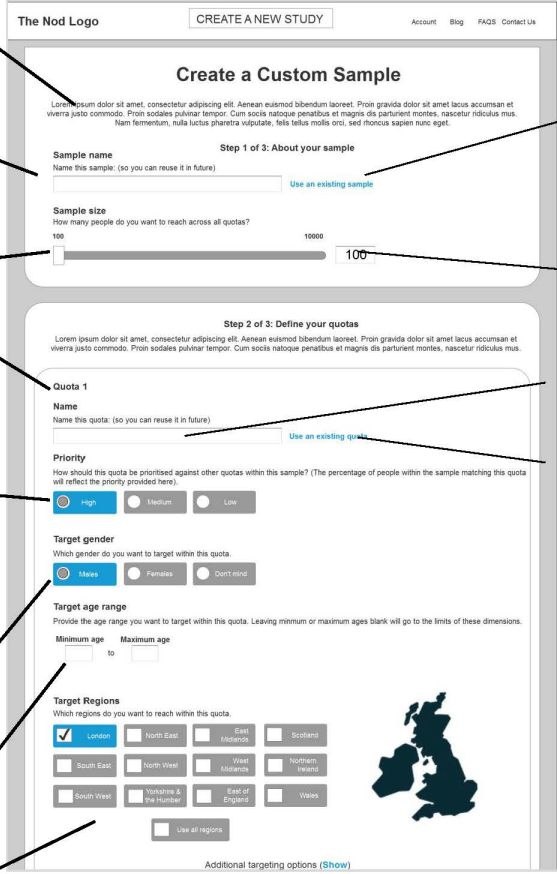
Upon arriving on Create a Custom sample users are asked to create Quota 1.

Users must set the priority for this quota. Priority will algorithmically determine the percentage of the sample that this quota is allocated to.

Gender is an option but included 'don't mind' option in case they are not bothered.

Users can input a minimum / maximum or age range.

Users can choose one or more regions. Selected regions are indicated on map (not shown here).



Clicking on use an existing sample will replace the text input field with a dropdown field within which users can select from the names of previous samples. When previous sample is chosen the rest of page updates accordingly. This link will change to 'create new sample'

Text input field updates as slider is moved whilst also allowing users to set the sample size without using the slider.

Users have the options of naming their quotas for future reference.

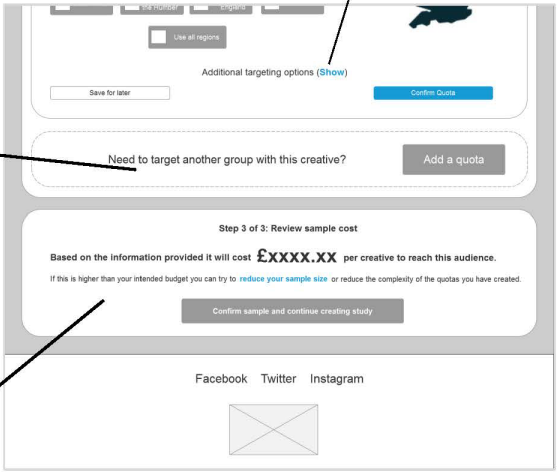
Works same as existing sample.

Users are prompted in case they want to create another quota. However option is inactive until current quota is confirmed.

Price is shown at bottom of the create a sample page. It updates whenever a new quota is added / edited or sample size is changed. Users are unable to continue creating a study until at least one quota is created.

Note: Unless I state otherwise then everything should look like the most recent visual designs.

More focused screener questions are hidden by default to reduce clutter. Clicking 'show' will reveal them.



The Nod Platform Design – Custom Quotas – Annotated Wireframes – Create a sample: Quota 1

For Discussion

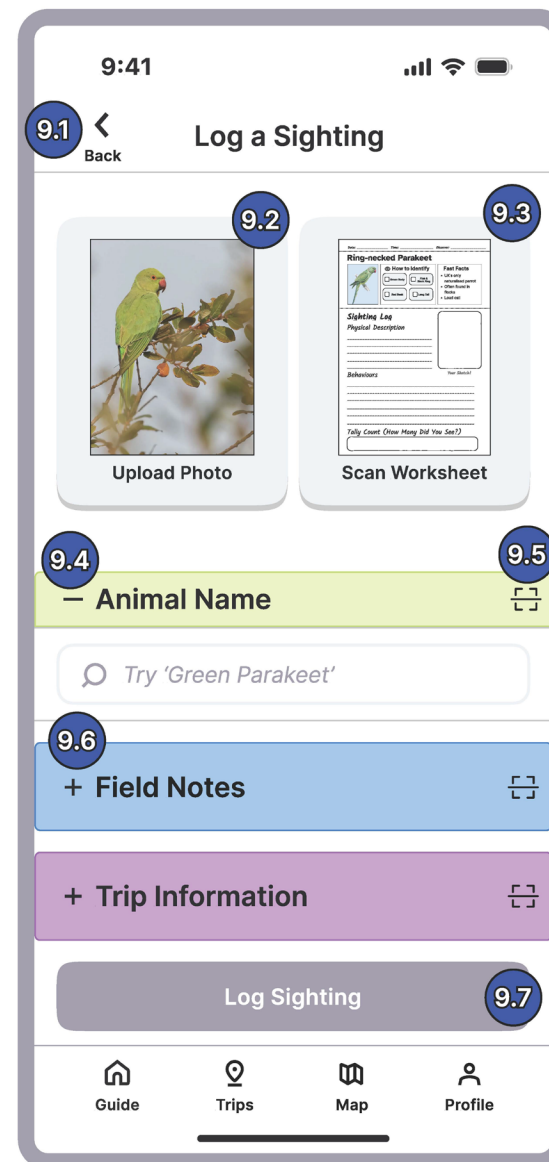
This document is not a project deliverable. It is being shared in order to discuss the design.

Version 0.1
09/08/2018
By: Stuart Scott
For: Ben Pask

Wireframe annotation examples

*There is some useful detail in these annotations.
Note also the unique wireframe ID and descriptive title*

W09: Log a Sighting



9.1 Tap "< Back" to return to previous page. If some of the form is filled out, user will need to confirm via popup that they understand their information will be lost before returning.

9.2 Tap anywhere on "Upload Photo" card to add a photo of the animal. User will be prompted to take a picture, add from their library, or cancel the operation. See W14, 14.5 for what added photo carousel looks like.

9.3 Tap anywhere on "Scan Worksheet" card to change to W11. (Change to W11)

9.4 Tap the "-" to collapse "Animal Name", which is expanded by default.

9.5 Tap the icon to scan handwritten notes (same flow as W11).

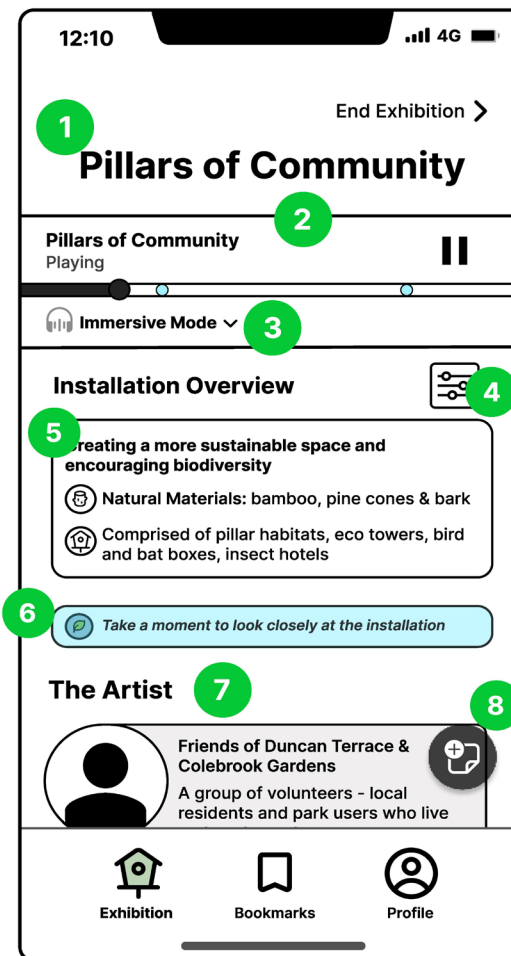
9.6 Tap the "+" to expand "Field Notes" section. See W10 for expanded sections. Both "Field Notes" and "Trip Information" are collapsed by default.

9.7 "Log Sighting" is grey to indicate that not enough information is provided. Once "Animal Name" is filled in, "Log Sighting" turns orange and can be tapped to successfully log a Sighting. If "Log Sighting" is tapped without "Animal Name" completed, a red caution message will appear above "Animal Name" saying "Add an Animal Name to Log Sighting". This will stay on the page with no time limit.

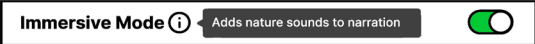
Wireframe annotation examples

*There is some useful detail
in these annotations,
although some aspects are
still a little vague.
Note the unique wireframe
ID and descriptive title*

W12. Art Installation Content Screen (Top)



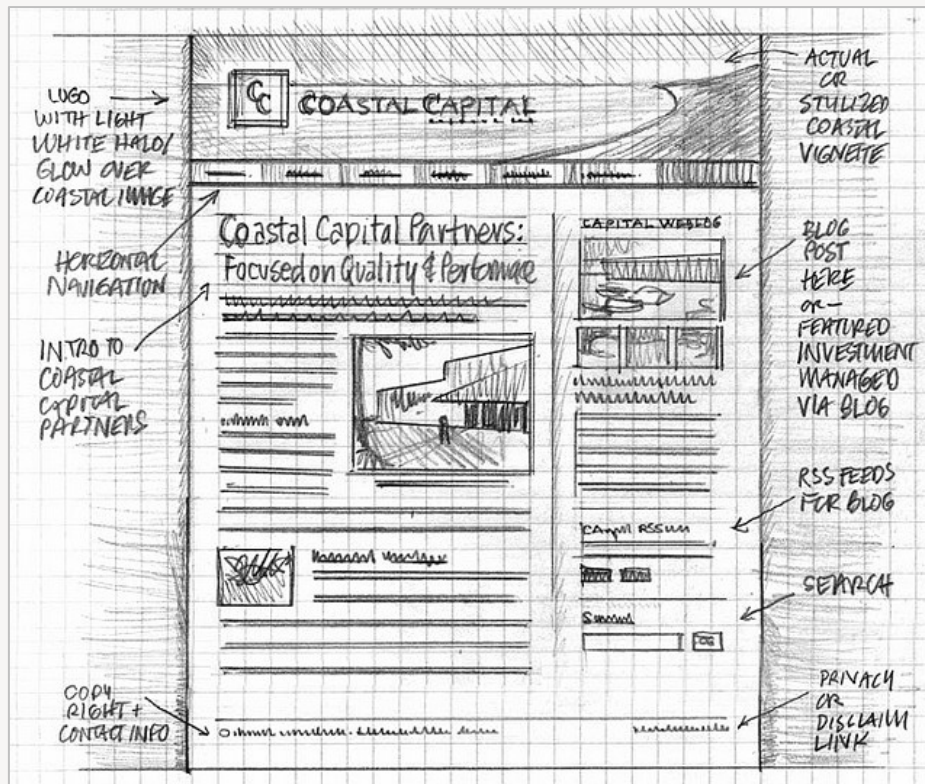
Annotations

- 1 Exhibition Content Screen**
Displays the art installation content during the active exhibition. Users scroll vertically to go through the content. This screen represents the audio-led, light overview experience. The content displayed changes based on the exhibition setup options selected.
- 2 Audio Bar**
Shows the audio narration for the installation when an audio-led experience is selected. The audio starts only when the user taps play. The audio bar remains fixed at the top of the screen as the user scrolls. Small blue markers in the audio progress bar indicate upcoming reflection pauses which last for a few seconds.
- 3 Immersive Mode**
Tapping this control expands and reveals a toggle to enable or disable immersive mode which layers ambient nature sounds under the narration audio. The icon appears grey when disabled and black when enabled.

- 4 Setup Adjustment Button**
Allows users to adjust exhibition preferences during the experience. Tapping opens a pop-up showing the user's current setup options and available alternatives. These changes happen immediately and the displayed exhibition content adapts to reflect the user's preferences.
- 5 Installation Overview Section**
Provides a brief introduction to the installation including its purpose, materials and key contextual information. This section aligns with the opening segment of the audio narration and adapts with the selected content depth.
- 6 Reflection Pause Prompt**
Timed prompts appear on screen at key moments in the exhibition to encourage users to pause and reflect on the installation. These prompts correspond to the pause markers in the audio progress bar.
- 7 Artist Section**
Displays the artist's name, image and a short background description. For audio-led experiences, a message from the artist is integrated into the main audio narration. This is presented as text for read-only experiences.
- 8 'Add Note' Floating Action Button**
Allows users to add personal notes about the installation at any point during the exhibition. Tapping the button navigates users to a screen where they can write a note and view notes shared by others (W12). Returning from the notes screen resumes the exhibition at the same installation.

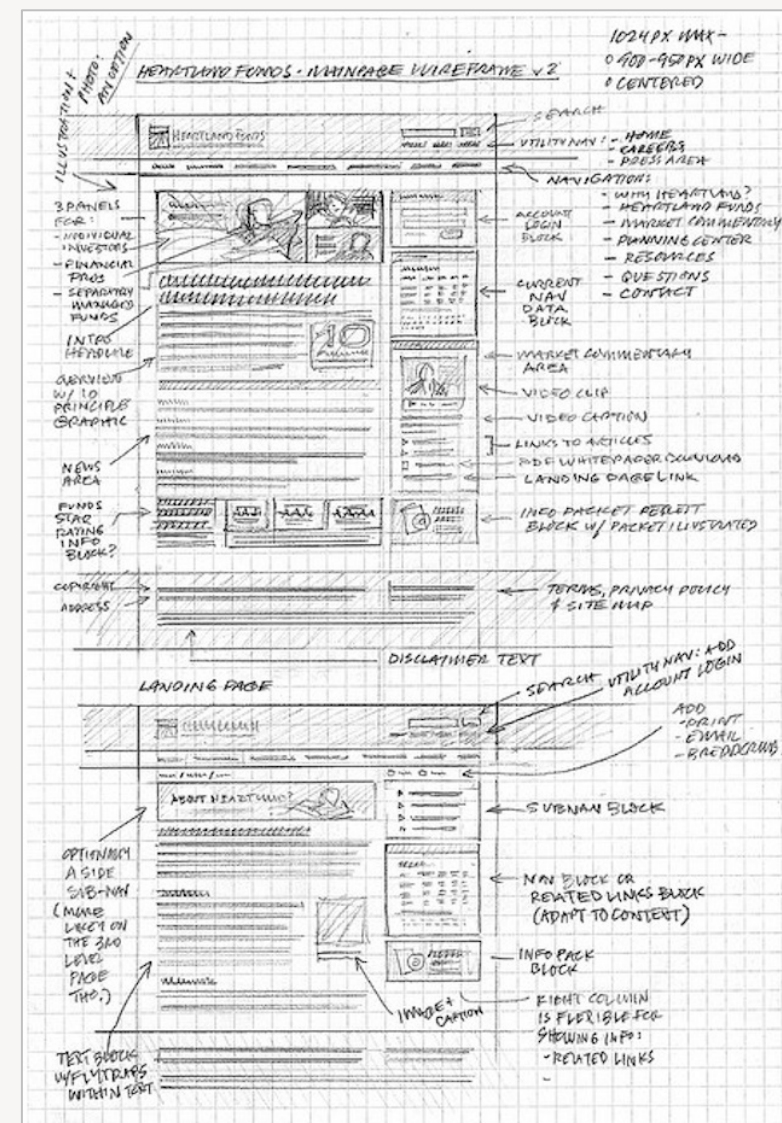
If a user is not logged in, tapping this button will prompt them to log in or sign up before leaving a note.

Early-stage wireframe sketch annotations examples

(Not suitable for the detailed design wireframes deliverable of the CW2 coursework—they're not detailed enough—but a nice illustration of some early planning)



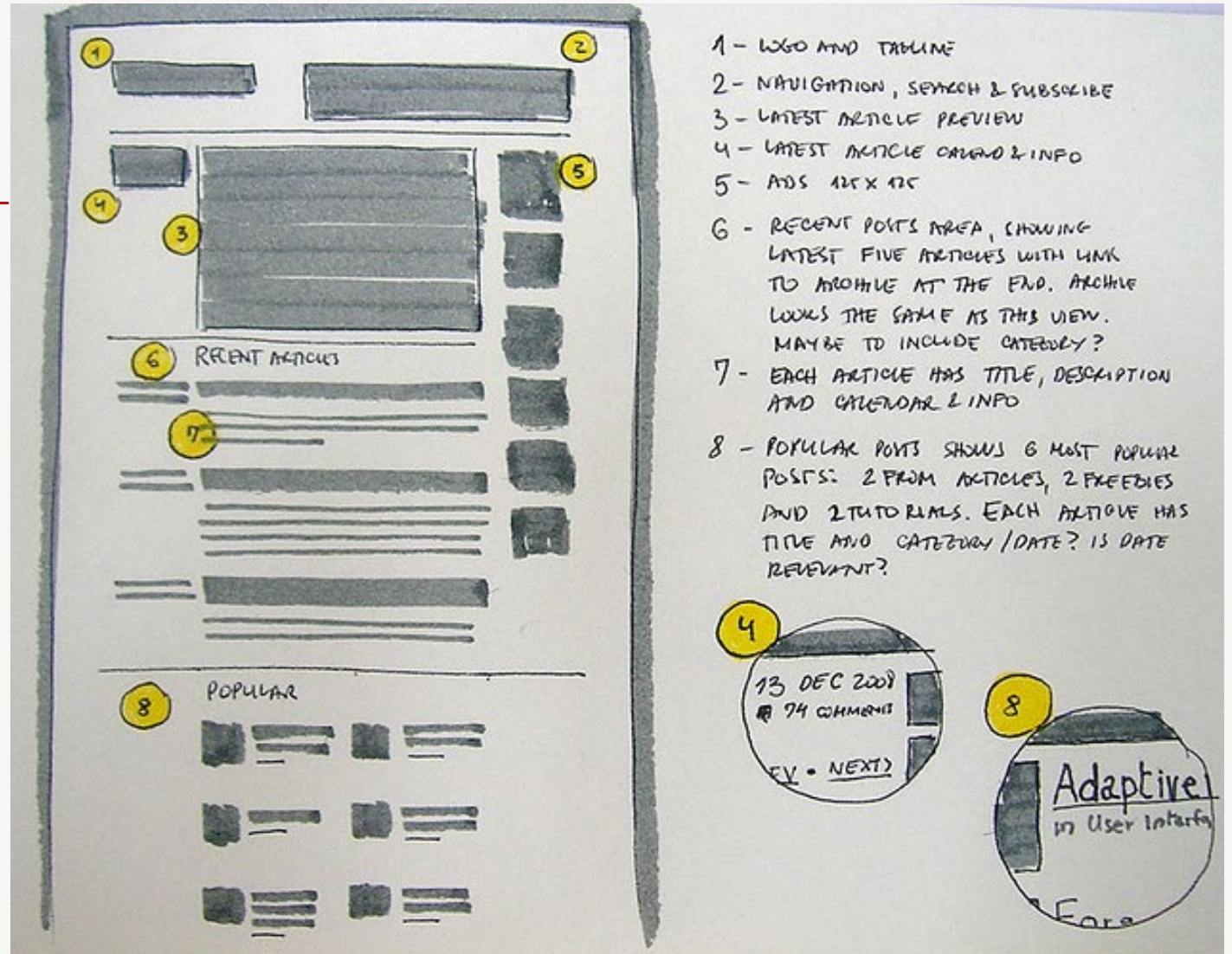
<https://www.flickr.com/photos/rohdesign/3307873748>



<https://www.flickr.com/photos/rohdesign/3307874546/>

Wireframe sketch annotations example

Not suitable for the CW2 coursework wireframes deliverable either, but an earlier step in your process might look like this.



Wireframes as **documentation**: Tips

- **Clarity is key**—these are communication artefacts.
- Choose an appropriate page orientation and size.
 - In digital files, make sure pages are oriented for reading.
 - Ideally stick to standard pages sizes, in case someone wants to print them out.
 - Provide enough room for annotations—one wireframe per page is often the norm.
- Provide a clear label for each screen design.
 - Unique ID and a descriptive title is a useful combination, e.g. “*W01. Homepage*”
 - You can then refer to the ID in other diagrams, e.g. user flow diagrams, etc.
- Ensure text is legible (except in parts where this isn’t important).
- Text and image quality should not degrade on zooming in.
- Include annotations, explaining content and functionality in appropriate detail.
- Make it clear how users would navigate between screens.
- Explanations should be detailed but succinct/concise.

CW2 coursework: Wireframes



- You must produce 6 annotated wireframes.
- They must support a particular journey / task—this must be represented in your user flow diagram deliverable.
- Choose wisely—ensure there is enough complexity to demonstrate your design skills to their greatest extent.
- This deliverable must be traceably grounded in your preceding work.
- The wireframes should be neat, with a professional look and feel.
- Annotations must be thorough and detailed—the annotated wireframes should serve as a specification of the interactive elements/features and behaviours of these screens.
- See the coursework brief for more details. Ask if unsure of something.

CW2 Wireframes watchpoints



Some* issues observed in previous submissions, that have impacted marks/results:

- ✗ Wireframes focused on generic features, rather than being specific to the proposed application.
- ✗ Wireframes do not clearly support a specific user journey or task, or the intended journey/task is unclear.
- ✗ Relationship to research, findings, persona etc not clear or traceable.
- ✗ Wireframes are missing titles.
- ✗ Annotations lack sufficient detail, are not specific enough, or not all relevant elements are annotated, leaving gaps in understanding. Use of vague, unsubstantiated language, e.g. 'intuitive', 'user-friendly'.
- ✗ Interaction logic, states or behaviour are missing, underdeveloped, or do not make sense.
- ✗ Inconsistencies in design, and behaviours, including across screens
- ✗ Poor presentation, e.g. image resolution poor (e.g. due to poor PDF production), untidy layout, inconsistent styling).
- ✗ Clear evidence of GenAI use, rather than students' own work.

* There are others, but these are some of the most common



Detailed Design

Some pointers

Filling in the details

- There is no magic formula!
- Refer to scenarios / user journey maps / storyboards / task models
- Use your personas. What tasks will users want or need to do?
- User flows help you to decide what screens need to be designed and what needs to be on them.
- Think about:
 - What information or functionality does the user need or expect?
 - How does the user trigger an action?
 - How does the system respond?

Sequence of interaction

- A user's journey through a system to complete a task is a **sequence of actions/interactions**.
- Identify all **relationships/dependencies** between activities:
 - Which items need to be completed before others;
 - Choices that are mutually exclusive.
- What might we need to accommodate via **alternative paths**?
- Be careful of having:
 - *One big screen*
 - Hard for users to pay full attention to the whole, and work out what to do next.
 - *Lots of small screens*—each having only one or two items.
 - This may be tiresome for users.
 - Think about what might logically be included on the same screen.

Things you might prioritise

You may not do detailed design for all aspects of the system, particularly to start with.

Consider prioritising:

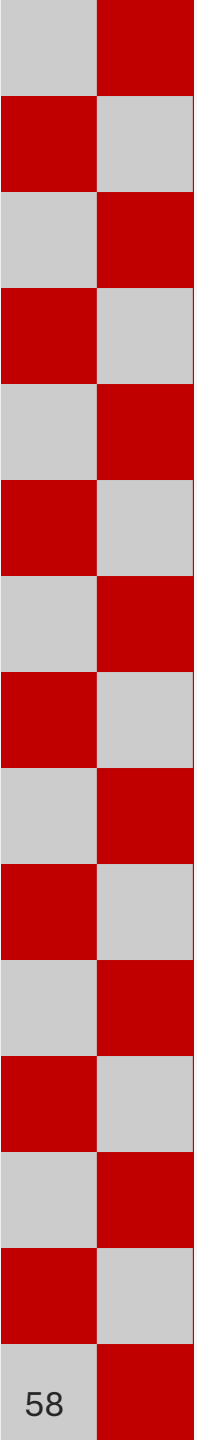
1. Novel or interesting interactions / journeys.
2. Any element that is used either repeatedly OR triggers key actions for the interaction.
3. Any operation that is known to be error-prone (from observation or previous evaluation) or ineffective.
4. Any operation that is known to demand particular effort or concentration, or may be more complex.

Iterate and evaluate

- Make sure to re-evaluate the design after each ‘iteration’
- Many short iterations are better than a few long ones where many design decisions have been made / many designs are ‘complete.’
- You will likely add further detail to previously “done” screens as you progress.
- Remember that you don’t need to complete all details at first.
 - Think about what to prioritise.
- Make sure that everything still works together – you can get from one screen to another.

Questions?





Design Principles and Guidelines

Design principles and guidelines

- Design principles and guidelines offer prescriptive advice.
- Many sets are available.
- Some are informed by cognitive psychology.
- Some are the result of empirical studies
 - e.g. Nielsen's heuristics.
- Some are based on experience and common sense.
- Some are constructed by organisations to ensure consistency across products, or that design is in line with certain values, e.g. inclusivity, sustainability...

(See section 8.4 in "Interaction Design")



Design principles

High-level rules to inform best practice

Design principles

- High level rules, often produced by experts, to inform best practice.
 - (e.g. Don Norman, Jakob Nielsen, Ben Schneiderman)
- “*accumulated wisdom of researchers and practitioners*” ([IxDF](#))
- Intended to guide designers' decisions when creating interfaces and interactive systemsCan also be used to evaluate the usability of systems.
- General guidance—can be applied to many contexts.
- Lots of examples here:
 - <https://principles.design>
 - <https://www.designprinciplesftw.com>

Interaction Design principles

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 - <https://principles.design>
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Jakob Nielsen's Heuristics

1. Visibility of System Status
2. Match Between the System and the Real World
3. User Control and Freedom
4. Consistency and Standards
5. Error Prevention
6. Recognition Rather than Recall
7. Flexibility and Efficiency of Use
8. Aesthetic and Minimalist Design
9. Help Users Recognize, Diagnose, and Recover from Errors
10. Help and Documentation

See: <https://www.nngroup.com/articles/ten-usability-heuristics>.

Jakob Nielsen's Heuristics (1/3)

1. Visibility of system status

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

2. Match between system and the real world

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.

3. User control and freedom

Users choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through extended dialogue. Support undo and redo.

4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

Jakob Nielsen's Heuristics (2/3)

5. Error prevention

Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

6. Recognition rather than recall

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

7. Flexibility and efficiency of use

Accelerators—unseen by the novice user—may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

Jakob Nielsen's **Heuristics** (3/3)

8. **Aesthetic and minimalist design**

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

9. **Help users recognize, diagnose, and recover from errors**

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

10. **Help and documentation**

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

Don Norman's 7 high-level design principles

1. **Discoverability:** Can users clearly see and understand the available options and controls?
2. **Feedback:** Do users receive immediate and clear information about what actions have been performed and the current state?
3. **Conceptual Model:** Can users form a good understanding of how the system operates, based on its design?
4. **Affordance:** Do appropriate means for interaction exist and are they supported by design?
5. **Signifiers:** Are there effective cues to ensure discoverability and communicate feedback clearly?
6. **Mappings:** Do controls and their movements intuitively correspond to their real-world effects?
7. **Constraints:** Are users guided to the correct actions and prevented from making errors?

See also: <https://uxdesign.cc/ux-psychology-principles-seven-fundamental-design-principles-39c420a05f84>

Affordances

- James Gibson 1977 *“affordances of the environment are what it offers the animal”*. —chair affords sitting; button affords pressing.
- Adopted by Norman, but often misunderstood/misinterpreted in the design community—seen as properties or even objects in their own right, rather than as intended.
- *“An affordance is a **relationship** between the properties of an object and the capabilities of the agent that determine just how the object could possibly be used.”* (Norman, 2013, Ch1)
- Affordances are what an object allows a particular user to do with it.
 - Not just about the object—the user is key to the relationship too.
- Some are perceivable, others are hidden.

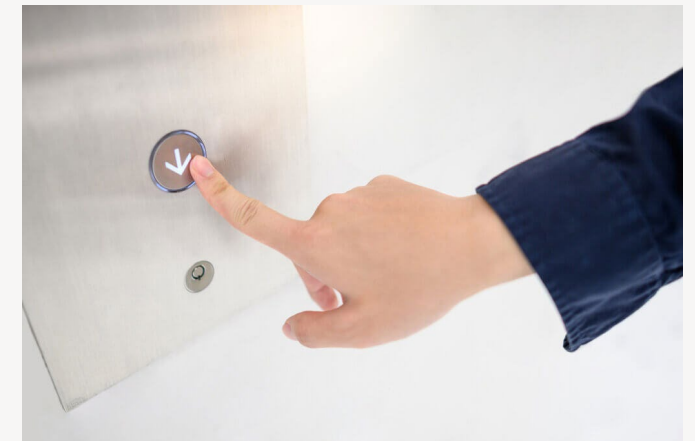
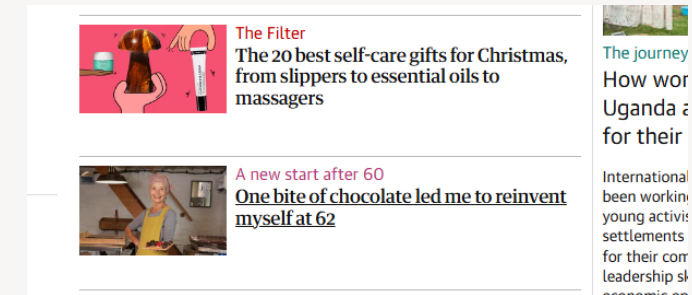


Signifiers

- Introduced by Norman, to address the issue with the term ‘affordances’.
- Visual cues or indicators in a design that signal what can be done where.

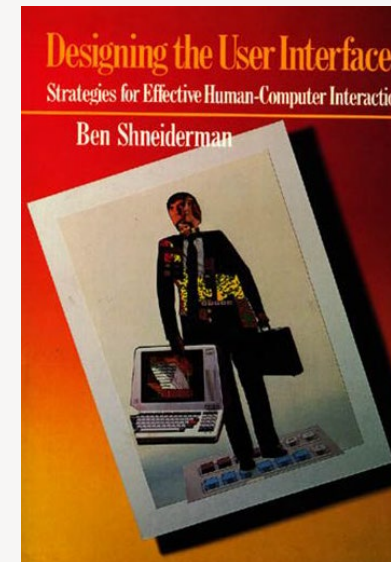
“any mark or sound, any perceivable indicator that communicates appropriate behaviour to a person” (Norman, 2019)

- Perceivable indicators of affordances. Help users discover possibility for interaction. e.g. push sign on a door; Label on a button.
- Communicate how to use a design.
- Can be deliberate, or accidental / incidental.
- Watch out for misleading with accidental signifiers i.e. implying something can be done that can’t.



Ben Shneiderman's 8 golden rules of interface design

1. Strive for consistency.
2. Seek universal usability.
3. Offer informative feedback.
4. Design dialogs to yield closure.
5. Prevent errors.
6. Permit easy reversal of actions.
7. Keep users in control.
8. Reduce short-term memory load.



Further explanation: <https://www.cs.umd.edu/users/ben/goldenrules.html>

<https://www.interaction-design.org/literature/article/shneiderman-s-eight-golden-rules-will-help-you-design-better-interfaces>

Some interaction design principles **applied**

- Be consistent.
- Reduce working memory load.
- Provide feedback to every user action.
- Make actions discoverable.
- Match the system to the real world.
- Prevent error.

Applying interaction design principles

- **Be consistent**

- Reduce working memory load
- Provide feedback to every user action
- Make actions discoverable
- Match the system to the real world.
- Prevent error

- Strive for consistency in:

- Menus & navigation
 - Color
 - Typography
 - Terminology
 - Layout
 - Capitalisation
 - Help screens
 - Sequences of actions... etc.
- Make use of conventions such as menu locations and structures, link styling, common terminology...

Applying interaction design principles

- Be consistent
- **Reduce working memory load**
- Provide feedback to every user action
- Make actions discoverable
- Match the system to the real world
- Prevent error
- Keep things simple.
 - Avoid clutter/distracting elements and unnecessary steps/complexity; Limit choices.
- Be consistent (previous slide), use conventions.
- Don't force people to process or remember a lot of things.
- Offload to the system, e.g. e.g. progress indicators to tell the user where they are in a process/task.

Applying interaction design principles

- Be consistent
- Reduce working memory load
- **Provide feedback for every user action**
- Make actions discoverable
- Match the system to the real world
- Prevent error
- Visual, Auditory, Haptic, Text...
- Should be prompt, perceivable, contextual & meaningful.
- Does the user know if the right/expected thing happened? (Norman's '*Gulf of Evaluation*')
- Examples: Sound, vibration or change in styling when something is clicked/pressed; Hover effects on clickable elements, including tooltips; Sound/vibration/text to notify of error/success; Specific patterns of vibration to indicate certain events/states; Error messages...

Applying interaction design principles

- Be consistent
- Reduce working memory load
- Provide feedback to every user action
- **Make actions discoverable**
- Match the system to the real world
- Prevent error
- Use visual cues of affordances.
 - e.g. through use of signifiers.
 - Element types, e.g. buttons, drop-down...
 - Styling, e.g. shadows
 - State change, e.g. hover effects, colour changes...
 - Labels, Tooltips
 - Iconography
- Be consistent.
- Follow conventions.
- Information Architecture principles.

Applying interaction design principles

- Be consistent
- Reduce working memory load
- Provide feedback to every user action
- Make actions discoverable
- **Match the system to the real world**
- Prevent error
- Use language familiar to the user and suitable for the context.
 - Avoid jargon.
- Match users' expectations, based on real world experiences and conventions
 - Consistency.
 - Natural mapping—correspondence between physical and digital; intuitive gestures; logical orders.
 - Familiar layouts and structures.
 - Colour and iconography choices.

Applying interaction design principles

- Be consistent.
- Reduce working memory load.
- Provide feedback to every user action.
- Make actions discoverable.
- Match the system to the real world.
- **Prevent error**
- Build in error prevention in online forms.
- Make things obvious or impossible by using constraints - gray out unavailable options.
- Use suggestions and autocomplete.
- Limit options.
- Real-time validation, with feedback, e.g. password setting.
- Guidance text / tooltips.
- Use confirmation dialogs.
- Provide Undo or Cancel buttons.

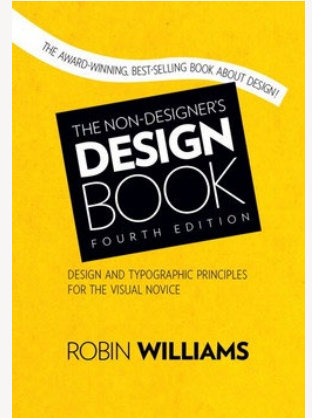
Visual design principles: **CRAP**

Contrast: Ensure good contrast between what you want people to pay attention to and other noise around it.
e.g. use colour contrast, image contrast, text size, image size.

Repetition: Keep consistency by repeating the same style across the design.
e.g. text size, link style, heading style, containers.

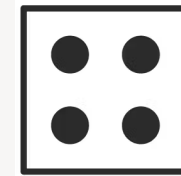
Alignment: Make sure everything is aligned and organised.
e.g. Text, images, form fields, bullet points.

Proximity: What belongs together stays together. Group elements which are related to each other (and put elements not related away from each other)



Visual design principles: IxDF

- Unity:** Use this to establish harmony between page elements, so they appear to belong together, and users aren't distracted by chaotic (e.g. misaligned) layouts.
- Gestalt:** Use these principles of how people perceive objects to guide how users interpret your design.
- Hierarchy:** Use placement, font, etc. to show importance.
- Balance:** Use this to distribute elements evenly.
- Contrast:** Use differences in color, etc. to accentuate elements.
- Scale:** Use this to emphasize elements to establish importance or depth.
- Dominance:** Use an object's size, color, etc. to make it stand out.



1. Unity



2. Gestalt



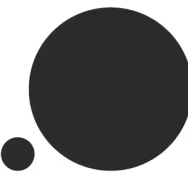
3. Hierarchy



4. Balance



5. Contrast



6. Scale



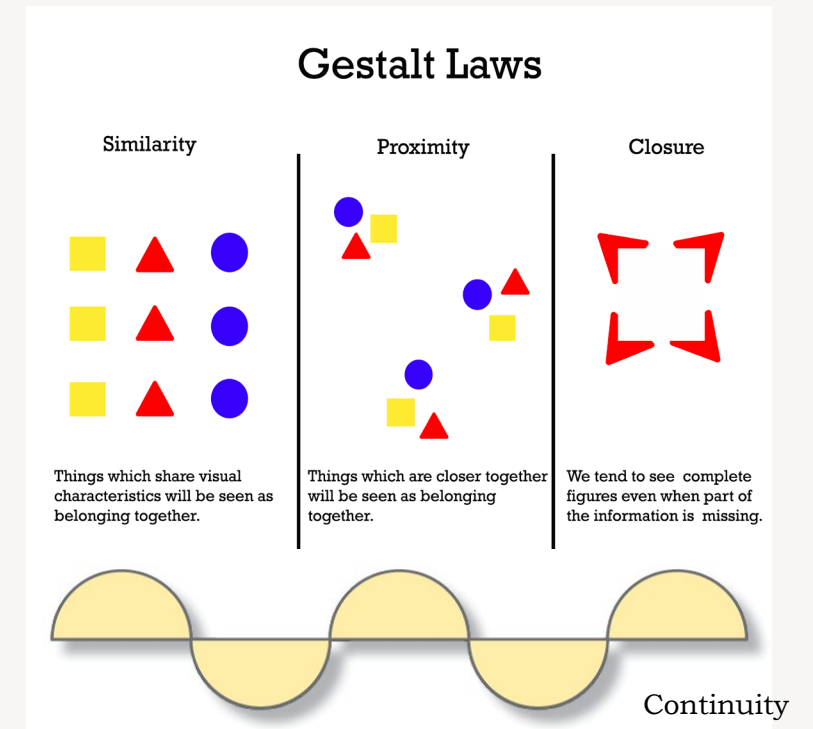
7. Dominance

From: <https://www.interaction-design.org/literature/topics/visual-design>

Gestalt laws of perception

Some most relevant for interface design:

- **Similarity**: similar objects tend to be grouped together.
- **Proximity**: things close in space or time will be perceived together.
- **Closure**: closed figures are perceived more easily; users tend to supply missing information.
- **Continuity**: people tend to perceive smooth, continuous patterns.
- **Pragnanz**: people perceive complex figures in simplest, clearest, most regular form.
- **Common region**: objects in a bounded area are perceived as a group.



Gestalt laws applied

- Figure / ground
- Proximity
- Common region
- Similarity

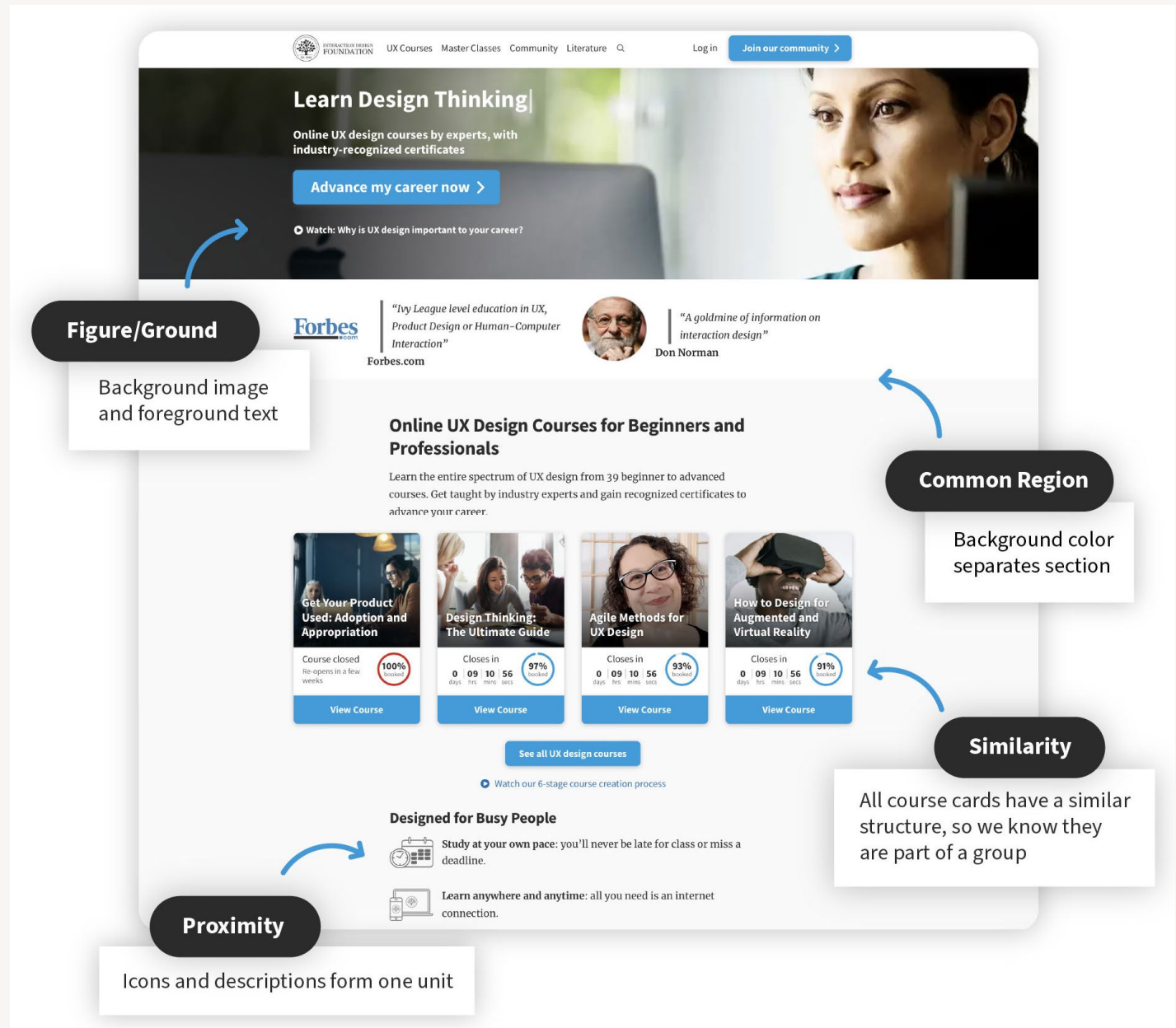


Image from <https://www.interaction-design.org/literature/topics/gestalt-principles>



Design guidelines (and beyond)

Specific **design** guidelines

- Often provided by software / hardware providers—to ensure that products designed for their platform are consistent and meet best practices for IxD.
- Describe how things should look or behave.
 - e.g. how to design screens, menus, controls or text for a specific platform.
 - [Android Design Guidelines](#)
 - [Designing for watchOS](#) (part of Apple's human interface guidelines)
- Can include general guidelines and considerations when designing for novel technologies.
- Designing for a particular platform or type of technology? Look for appropriate guidelines.

Example: Apple Watch best practices



Great Apple Watch experiences are streamlined and specialized, and integrate the platform and device capabilities that people value most. To help your experience feel at home in watchOS, prioritize the following ways to incorporate these features and capabilities.

- *Support quick, glanceable, single-screen interactions that deliver critical information succinctly and help people perform targeted actions with a simple gesture or two.*
- *Minimize the depth of hierarchy in your app's navigation, and use the Digital Crown to provide vertical navigation for scrolling or switching between screens.*
- *Personalize the experience by proactively anticipating people's needs and using on-device data to provide actionable content that's relevant in the moment or very soon.*
- *Use complications to provide relevant, potentially dynamic data and graphics right on the watch face where people can view them on every wrist raise and tap them to dive straight into your app.*
- *Use notifications to deliver timely, high-value information and let people perform important actions without opening your app.*
- *Use background content such as color to convey useful supporting information, and use materials to illustrate hierarchy and a sense of place.*
- *Support Siri to help people access shortcuts on the Siri watch face.*
- *Design your app to function independently, complementing your notifications and complications by providing additional details and functionality.*

<https://developer.apple.com/design/human-interface-guidelines/designing-for-watchos>

Specific design guidelines: **pros & cons**

Advantages:

- Consistency with user expectations
- Improved usability & learnability
- Faster design & development
- Perceived quality
- Better integration with OS features

Disadvantages:

- Can constrain design possibilities and creativity.
- Not all guidelines suit all use cases
- May be harder to maintain a unified cross-platform experience

Design patterns

- Reusable solutions for common design problems (e.g. scroll to refresh, use eye icon show/hide password, etc).
 - No need to solve something that has already been solved.
 - Users are often familiar with widely used patterns—**usability advantage!**
- Designers choose and adapt a pattern (solution) to fit their specific context.
- Numerous public pattern libraries for interaction design including: <http://ui-patterns.com> and <https://www.checklist.design/>
- Organisations may have their own internal pattern libraries, e.g:
 - GDS: <https://design-system.service.gov.uk/patterns/>
 - BBC: <https://www.bbc.co.uk/gel/guidelines/category/design-patterns>

Example: BBC GEL design pattern

Design Patterns	▼
Contents	▲
<u>Overview</u>	
<u>How it works</u>	
<u>Rules</u>	
<u>Accessibility</u>	
<u>Variations</u>	
<u>Pattern in action</u>	

Design Patterns

Patterns are a simple, reusable solution to a design problem.

FILTER: All Web App TV



Accordion

An accordion is a vertical list of headers, users interact to reveal or hide content within them. They reduce page length and scrolling.

v1 13 Jul 2016

How it works

At its most granular form an accordion can be broken down into an individual component. This can be referred to as a drawer. This is made up of a drawer header and content panel.

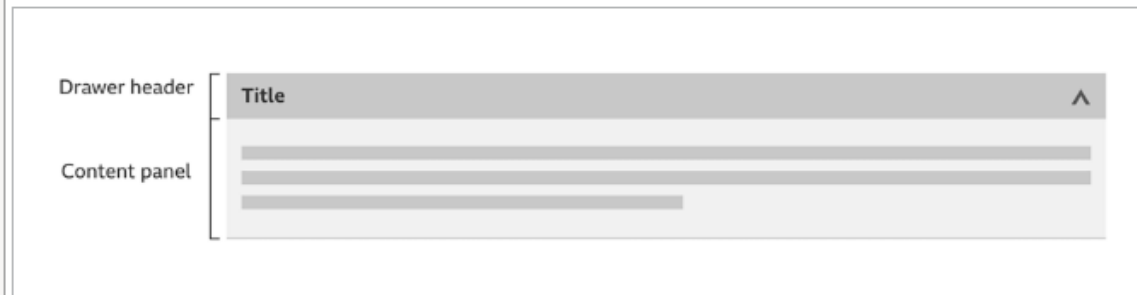
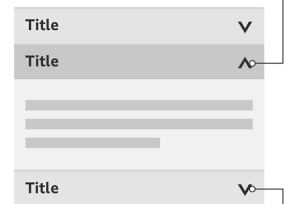


Fig 2 An example of a drawer header and content panel

Using an accordion to group content into individual drawers leads to shortened pages and reduced scrolling. It's in your control to either open or close a drawer.

✓ Correct arrow alignment

Arrow indicates drawer will collapse upwards



Arrow indicates drawer will expand downwards

✗ Incorrect arrow alignment



For more information about this pattern see: <https://www.bbc.co.uk/gel/features/accordion>

Design systems

- Often created by large companies / organisations.
- Core elements, patterns, reusable components, styles, guidelines, sometimes principles.
- May be proprietary, or open source.

Examples:

- Apple HIG: <https://developer.apple.com/design/human-interface-guidelines>
- Atlassian DS: <https://atlassian.design>
- BBC GEL: <https://www.bbc.co.uk/gel>
- Google Material Design: <https://m2.material.io>
- GOV.UK DS: <https://design-system.service.gov.uk>
- NHS DS: <https://service-manual.nhs.uk/design-system>
- Microsoft Fluent: <https://fluent2.microsoft.design>

Questions?



Session 8 Summary

- Wireflows (hybrid of wireframes + user flow) can be used to show transitions.
- Wireframes present a simplified view of screen structure and layout. They document design decisions and communicate the structure and behaviour of an interface. Annotations aid understanding.
- Detailed design involves ‘filling in the details’. There is no magic formula, but a series of important decisions, and, ideally, iteration.
- Design principles and guidelines can guide design work and help to ensure good usability and a positive user user experience.
- By adopting visual design principles we can help users to intuitively understand the structure of an interface and the functionality it offers.
- Design patterns and design systems also help designers of interactive technology.

Module Evaluation Questionnaire

<https://city.surveys.evasysplus.co.uk>

Closes Sunday 12th April

